
Efficient Inference for Noisy LLM-as-a-Judge Evaluation

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Abstract

Large language models (LLMs) are increasingly used as automatic evaluators of generative AI outputs, a paradigm often referred to as “LLM-as-a-judge.” In practice, LLM judges are imperfect predictions for the underlying truth and can exhibit systematic, non-random errors. Two main approaches have recently been proposed to address this issue: (i) direct measurement-error correction based on misclassification models and (ii) surrogate-outcome approaches, which correct bias by calibrating prediction residuals on a small set of gold-standard human labels. In this paper, we study these approaches side by side for mean parameters (e.g., average benchmark scores or pairwise win rates). Leveraging semiparametric efficiency theory, we provide a unified comparison of RG-style correction, PPI, PPI++, EIF-based calibration, and joint MLE, clarify when EIF and optimally tuned PPI++ coincide in the binary setting, and use this lens to guide extensions beyond binary outcomes.

1. Introduction

Automated model evaluation is an essential yet challenging task in machine learning, especially in the era of generative artificial intelligence (AI), where scalability produces large volumes of outputs to be labeled and introduces diverse output formats ranging from numerical predictions and categorical labels to open-ended paragraphs such as reasoning traces and mathematical proofs, all of which are not adequately captured by traditional metrics such as mean squared error or misclassification rates. Moreover, studies now increasingly leverage large language models (LLMs)

to generate substantial quantities of new questions *and* the corresponding answers, for which running every question-answer pair through human verification is infeasible.

Motivated by recent advances in LLMs, a growing body of work proposes using LLMs as automatic “judges” to evaluate the outputs of other models—for example, by determining whether a model’s answer is correct, assigning a quality score, or selecting the preferred response between two candidates, a practice known as “LLM-as-a-judge”. This approach promises scalable and low-cost evaluation, but it also introduces a new source of error: the LLM judge is an imperfect proxy for human judgment. At a high level, LLM-as-a-judge workflows typically rely on (i) a large *evaluation set* on which *only* LLM judgments are available; and (ii) a smaller *calibration set* on which both human labels and LLM judgments are observed.

This setup presents a challenge: relying solely on the calibration set leads to high-variance evaluation, whereas relying on the evaluation set labels leads to biased evaluation, as LLMs are known to produce homogenized judgments and to avoid extreme categories. Recent work has therefore asked the key question: *How can we optimally combine these two datasets to construct valid, efficient estimates with calibrated uncertainties for evaluation metrics?*

Statistically, LLM-as-a-judge evaluation can be framed as a classical measurement-error problem. Consider the binary-outcome setting where the human label Y represents the latent truth and the LLM judgment $\hat{Y}$ is a noisy surrogate. We observe paired data $(Y_i, \hat{Y}_i)$ for a small calibration subset, while for most of the evaluation set only $\hat{Y}_i$ is available. A long-standing approach in this literature is to model the misclassification mechanism $\Pr(\hat{Y} | Y)$ and estimate the true accuracy using *direct measurement-error correction*, such as Rogan–Gladen estimators or confusion-matrix inversion (Rogan & Gladen, 1978; Fuller, 2009). Another school of thought, including *prediction-powered inference* (PPI) (Angelopoulos et al., 2023a) and related work (Wang et al., 2020; Miao et al., 2025; Salerno et al., 2025; Ji et al., 2025; Egami et al., 2023; Rister Portinari Maranca et al., 2025), treats the LLM judge as a generic surrogate outcome for Y and uses the calibration set to correct the *average bias* of $\hat{Y}$. This approach yields valid statistical inference without explicit estimation of the misclassification model.

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Despite progress in both methodological traditions, there has been little direct comparison of their statistical validity and efficiency. This gap likely reflects the distinct cultures and communities in which these approaches are typically applied: measurement-error models have recently been used primarily in medical settings, such as estimating COVID-19 and cause-of-death prevalence under imperfect measurements (Rosin et al., 2023; González et al., 2017; Fiksel et al., 2022), whereas PPI-style estimators are more commonly motivated by modern ML/AI applications. Some recent exceptions have applied measurement-error corrections in LLM-as-a-judge settings (Lee et al., 2025), studied upper bounds on the efficiency gains achievable by such judges (Dorner et al., 2024), and independently proposed similar calibration procedures (Landesberg, 2025).

We study these two approaches for mean parameters and use semiparametric efficiency theory as a common lens. Our EIF construction should be viewed as the scalar/binary specialization of the same residual-correction perspective that underlies recent RePPI-style and generalized PPI work (Ji et al., 2025; Gronsbell et al., 2024; Zou et al., 2026), with the additional benefit that in this LLM-as-a-judge setting the comparison to direct misclassification correction can be written explicitly. EIF theory for mean estimation of Y with surrogate $\hat{Y}$ shows that the optimal estimator depends on $\mathbb{E}[Y \mid \hat{Y}]$. This perspective reveals that popular calibration-based estimators approximate $\mathbb{E}[Y \mid \hat{Y}]$ using different functional forms: PPI implicitly uses the identity $Y \approx \hat{Y}$; PPI++ uses a one-parameter linear map $Y \approx \lambda \hat{Y}$; and RePPI replaces these parametric forms with flexible machine learning. In the binary-outcome setting, PPI++ with optimal λ coincides with the EIF-based estimator, whereas for general outcomes this need not hold. These insights connect to the literature on efficient inference for missing data, especially missing completely at random (Chen et al., 2008; Robins & Rotnitzky, 1995). They also clarify prior efficiency results: Dorner et al. (2024) show that PPI++ attains the semiparametric efficiency bound for mean estimation with linear form, while Xu et al. (2025) show more generally that PPI++ is suboptimal unless $\mathbb{E}[Y \mid \hat{Y}]$ is linear.

Our contribution is therefore primarily one of unification, explicit comparison, and practical guidance. (1) *Empirically*, we benchmark RG-style correction, PPI, PPI++, EIF, and joint MLE in shared LLM-as-a-judge settings and show a consistent coverage-width ordering across simulations and real data. (2) *Theoretically*, we prove that the EIF estimator is asymptotically equivalent to optimally tuned PPI++ in the binary case, and that standard PPI remains less efficient while still improving substantially on direct inversion-based correction. (3) *Practically*, this shared lens yields concrete guidance on when simple linear calibration is enough, when flexible calibration matters, and how these

efficiency gains should be interpreted relative to robustness under distribution shift.

Finally, while this work focuses on guarantees of inferential validity for mean parameters, there is a growing body of research aimed at improving LLM-as-a-judge scores ($\hat{Y}$) so that $\hat{Y} \approx Y$ across a diverse set of tasks and LLM models (Tian et al., 2025; Liu et al., 2024; Shi et al., 2024; Sahoo et al., 2025; Schroeder & Wood-Doughty, 2024; Li et al., 2025). We view this line of work as complementary: our goal is to provide guarantees regardless of the quality of $\hat{Y}$, while improved calibration of $\hat{Y}$ can further reduce variance for our estimator.

Conflict of Interest Disclosure. The authors declare no financial conflicts of interest.

2. Calibration Methods for Binary Outcomes

2.1. Problem Setup

We stylize the LLM-as-a-judge evaluation as follows: for $i = 1, \dots, N$, we observe i.i.d. pairs

$$Y_i \sim \text{Bern}(\theta), \quad \hat{Y}_i \in \{0, 1\} \quad (1)$$

denoting the ground truth (e.g., from a human expert) and the LLM-as-a-judge label, respectively. Our parameter of interest is the *mean human outcome*

$$\theta = \mathbb{E}[Y] = \Pr(Y = 1). \quad (2)$$

Throughout the binary section, we write $q_1 = \Pr(\hat{Y} = 1 \mid Y = 1)$ for judge sensitivity and $q_0 = \Pr(\hat{Y} = 0 \mid Y = 0)$ for judge specificity. As a concrete example, if the goal is to compare two responses R_1 and R_2 produced by different LLMs, let $Y_i = 1$ indicate that a human annotator prefers R_1 over R_2 for prompt i (0 otherwise). The estimand θ then represents the *human win rate* of R_1 over R_2 .

The full sample of size N is partitioned into two *independent*, disjoint subsets: a **test set** of size n , in which only the surrogate outcome $\hat{Y}_i$ is observed, and a **calibration set** of size $m = N - n$, in which both the true outcome Y_j and the surrogate $\hat{Y}_j$ are observed. This setup corresponds exactly to a *missing completely at random* (MCAR) mechanism, where the true outcome Y_i is missing for observations in the test set (Robins & Rotnitzky, 1995; Little & Rubin, 2002), because the partition into calibration versus test sets is generated by an independent Bernoulli assignment that does not depend on either Y_i or $\hat{Y}_i$. For asymptotic analysis, we consider a regime in which both m and n diverge with $\frac{n}{m} \rightarrow \gamma_1 \in (0, \infty)$. That is, the proportion of observations in the calibration set converges to a fixed constant.

Denote the proportion of LLM-judged positives by

$$\hat{\theta}_{\text{naive}} = \frac{1}{n} \sum_{i=1}^n \hat{Y}_i. \quad (3)$$

When the judge is imperfect (i.e., $q_0 + q_1 < 2$), we have that $\mathbb{E}[\hat{\theta}_{\text{naive}}] = \theta + (q_0 + q_1 - 2)\theta + (1 - q_0)$, which leads to biased point estimates, incorrect variance estimates, and invalid inference. We therefore review several calibration-based/debiasing estimators: the Rogan–Gladen (RG) estimator (Rogan & Gladen, 1978; Lee et al., 2025), prediction-powered inference (PPI) (Angelopoulos et al., 2023a), and its power-tuned variant (PPI++) (Angelopoulos et al., 2023b).

2.2. Rogan–Gladen Estimator

The Rogan–Gladen estimator accounts for the sensitivity and specificity of the LLM judge,

$$q_1 = \Pr(\hat{Y} = 1 \mid Y = 1), \quad q_0 = \Pr(\hat{Y} = 0 \mid Y = 0). \quad (4)$$

Let $p = \Pr(\hat{Y} = 1)$ denote the marginal probability that the LLM calls an instance positive. We have that

$$p = q_1\theta + (1 - q_0)(1 - \theta) = (q_0 + q_1 - 1)\theta + (1 - q_0). \quad (5)$$

Solving (5) for θ yields $\theta = \frac{p + q_0 - 1}{q_0 + q_1 - 1}$. In practice, p , q_0 , and q_1 are unknown and estimated from the data: $\hat{p} = \frac{1}{n} \sum_{i=1}^n \hat{Y}_i$. On the calibration set, write $m_1 = \sum_{j=1}^m \mathbf{1}\{Y_j = 1\}$ and $m_0 = m - m_1$ and define $\hat{q}_1 = \frac{\sum_{j=1}^m \mathbf{1}\{\hat{Y}_j = 1, Y_j = 1\}}{m_1}$, $\hat{q}_0 = \frac{\sum_{j=1}^m \mathbf{1}\{\hat{Y}_j = 0, Y_j = 0\}}{m_0}$. The Rogan–Gladen estimator for θ is

$$\hat{\theta}_{\text{RG}} = \frac{\hat{p} + \hat{q}_0 - 1}{\hat{q}_0 + \hat{q}_1 - 1}. \quad (6)$$

This is the classic prevalence estimator under measurement error (Rogan & Gladen, 1978; Lang & Reiczigel, 2014) and the same estimator proposed in Lee et al. (2025).

Proposition 2.1 (Asymptotic distribution of $\hat{\theta}_{\text{RG}}$ in (6)). *Assume the test and calibration samples are independent and $q_0 + q_1 - 1 \neq 0$ with q_0 and q_1 defined in (4). With $N = n + m \rightarrow \infty$ and $n/m \rightarrow \gamma_1$, we have that*

$$\sqrt{N}(\hat{\theta}_{\text{RG}} - \theta) \xrightarrow{d} \mathcal{N}(0, V_{\text{RG}}),$$

where $V_{\text{RG}} = \frac{(1 + \gamma_1)[p(1 - p) + \gamma_1\{(1 - \theta)q_0(1 - q_0) + \theta q_1(1 - q_1)\}]}{\gamma_1(q_0 + q_1 - 1)^2}$.

2.3. Prediction-Powered Inference (PPI)

Prediction-powered inference (Angelopoulos et al., 2023a) corrects the bias of $\hat{\theta}_{\text{naive}}$ using the difference in means on calibration set. Since we have that $\mathbb{E}[\hat{Y}] = p$, $\mathbb{E}[\hat{Y} - Y] = \Delta = p - \theta$, it’s natural to consider the following estimator:

$$\hat{\theta}_{\text{PPI}} = \hat{p} - \frac{1}{m} \sum_{j=1}^m (\hat{Y}_j - Y_j). \quad (7)$$

Proposition 2.2. *Under model (1) with independent test and calibration sets, we have that (i) $\hat{\theta}_{\text{PPI}}$ is unbiased for any n, m ; and (ii) $\hat{\theta}_{\text{PPI}}$ is asymptotically normal with variance $V_{\text{PPI}} = \left(\frac{1 + \gamma_1}{\gamma_1}\right) \{p(1 - p) + \gamma_1[(1 - \theta)(1 - q_0) + \theta(1 - q_1) - (\theta - p)^2]\}$.*

2.4. Power-Tuned PPI (PPI++)

PPI++ generalizes (7) by introducing a parameter λ :

$$\hat{\theta}_{\text{PPI++}}(\lambda) := \frac{1}{m} \sum_{j=1}^m Y_j + \lambda \left(\frac{1}{n} \sum_{i=1}^n \hat{Y}_i - \frac{1}{m} \sum_{j=1}^m \hat{Y}_j \right). \quad (8)$$

It’s easy to see that (8) remains unbiased for any choice of λ that is independent of the Y_i ’s, but different choices lead to different asymptotic variances (and therefore different power), as detailed in Proposition 2.3.

Proposition 2.3. *Assume λ is fixed, under model (1) with independent test and calibration samples, we have that (i) $\mathbb{E}[\hat{\theta}_{\text{PPI++}}(\lambda)] = \theta$ for any n, m ; (ii) $\sqrt{N}(\hat{\theta}_{\text{PPI++}}(\lambda) - \theta) \xrightarrow{d} \mathcal{N}(0, V_{\text{PPI++}}(\lambda))$,*

$$V_{\text{PPI++}}(\lambda) = \frac{1 + \gamma_1}{\gamma_1} \left[\gamma_1 \theta(1 - \theta) + \lambda^2(1 + \gamma_1)p(1 - p) - 2\lambda \gamma_1 \theta(q_1 - p) \right]; \quad (9)$$

and (iii) $V_{\text{PPI++}}(\lambda)$ is minimized at $\lambda^* = \frac{\gamma_1 \theta(q_1 - p)}{(1 + \gamma_1)p(1 - p)}$.

2.5. Maximum Likelihood Estimator

Another natural candidate in the binary case is the *maximum likelihood estimator* under the misclassification model for $(Y, \hat{Y})$. In contrast to RG, which uses plug-in estimators for sensitivity and specificity on the calibration sample, we could jointly fit (θ, q_0, q_1) by maximizing the likelihood implied by the model across the test and calibration samples. Specifically, under model (1), the test observations satisfy $\Pr(Y = 1, \hat{Y} = 1) = \theta q_1$, $\Pr(Y = 1, \hat{Y} = 0) = \theta(1 - q_1)$, $\Pr(Y = 0, \hat{Y} = 1) = (1 - \theta)(1 - q_0)$, and $\Pr(Y = 0, \hat{Y} = 0) = (1 - \theta)q_0$.

Likelihood and estimating equations. Let $O_i = (R_i, Y_i, \hat{Y}_i)$ for $i = 1, \dots, n + m$, where $R_i \in \{0, 1\}$ indicates whether Y_i is observed ($R_i = 1$ for calibration and $R_i = 0$ for test). Define $S = \sum_{i=1}^{n+m} (1 - R_i)\hat{Y}_i$ and $N_{y\hat{y}} = \sum_{i=1}^{n+m} R_i \mathbf{1}\{Y_i = y, \hat{Y}_i = \hat{y}\}$. Then the log-likelihood (up to constants) can be written as

$$\begin{aligned} \ell(\theta, q_0, q_1) = & S \log p + (n - S) \log(1 - p) \\ & + N_{11} \log(\theta q_1) + N_{10} \log(\theta(1 - q_1)) \\ & + N_{01} \log((1 - \theta)(1 - q_0)) \\ & + N_{00} \log((1 - \theta)q_0). \end{aligned} \quad (10)$$

The MLE is $(\hat{\theta}_{\text{MLE}}, \hat{q}_{0, \text{MLE}}, \hat{q}_{1, \text{MLE}}) = \arg \max_{\theta, q_0, q_1 \in (0, 1)} \ell(\theta, q_0, q_1)$, equivalently a solution to $\nabla_{\eta} \ell(\eta) = 0$ with $\eta = (\theta, q_0, q_1)^\top$. In practice, we

solve this numerically (e.g., Newton–Raphson), requiring $q_0 + q_1 \neq 1$ for identifiability.

Proposition 2.4 (Asymptotic normality and efficiency of the MLE). *Suppose $N = n + m \rightarrow \infty$ with $n/m \rightarrow \gamma_1 \in (0, \infty)$, and define $a = q_0 + q_1 - 1$, $v_0 = q_0(1 - q_0)$, $v_1 = q_1(1 - q_1)$, and $b = (1 - \theta)v_0 + \theta v_1$. We have that $\sqrt{N}(\hat{\eta}_{\text{MLE}} - \eta) \xrightarrow{d} \mathcal{N}(0, I_{\gamma_1}(\eta)^{-1})$, where $I_{\gamma_1}(\eta) = \frac{\gamma_1}{1 + \gamma_1} I_{\text{test}}(\eta) + \frac{1}{1 + \gamma_1} I_{\text{cal}}(\eta)$ with*

$$I_{\text{test}}(\eta) = \frac{1}{p(1-p)} \begin{pmatrix} a^2 & -a(1-\theta) & \theta a \\ -a(1-\theta) & (1-\theta)^2 & -\theta(1-\theta) \\ \theta a & -\theta(1-\theta) & \theta^2 \end{pmatrix},$$

and

$$I_{\text{cal}}(\eta) = \text{diag}\left(\frac{1}{\theta(1-\theta)}, \frac{1-\theta}{v_0}, \frac{\theta}{v_1}\right).$$

Consequently, $\sqrt{N}(\hat{\theta}_{\text{MLE}} - \theta) \xrightarrow{d} \mathcal{N}(0, [I_{\gamma_1}(\eta)^{-1}]_{11})$, where

$$[I_{\gamma_1}(\eta)^{-1}]_{11} = \frac{(1 + \gamma_1)\theta(1-\theta)\{p(1-p) + \gamma_1 b\}}{p(1-p) + \gamma_1 b + \gamma_1 a^2 \theta(1-\theta)}. \quad (11)$$

3. A Principled Efficiency Lens via Efficient Influence Functions

We now have multiple asymptotically unbiased estimators for $\theta = \mathbb{E}[Y]$ (RG, PPI, PPI++, and the MLE). This motivates the natural question:

What is the smallest achievable asymptotic variance under a given statistical model, and which estimators attain it?

We answer this question using semiparametric efficiency theory in statistics (see Bickel et al. (1993); Van der Vaart (2000) and references therein for a detailed treatment). Consider i.i.d. $Z_1, \dots, Z_N \sim P$ and let $\theta = \theta(P)$ be the target parameter. An estimator $\hat{\theta}$ is *regular and asymptotically linear* if there exists a mean-zero function $\phi_P(Z)$ such that $\hat{\theta} - \theta(P) = \frac{1}{N} \sum_{i=1}^N \phi_P(Z_i) + o_p(N^{-1/2})$, where $o_p(N^{-1/2})$ denotes a remainder term that is negligible relative to $N^{-1/2}$ under P . Any such ϕ_P is called an *influence function* (IF) for $\hat{\theta}$, and a direct application of the central limit theorem yields that $\hat{\theta}$ is asymptotically normal.

Among all IFs, the *efficient influence function* (EIF) for θ under P , when it exists, is the unique IF with the smallest asymptotic variance. This variance is the *semiparametric efficiency bound*. An estimator with the EIF as its IF attains this bound and is *semiparametrically efficient*.

Now return to our joint model for calibration and test samples $(Y, \hat{Y}, R) \sim P$, where $Y \in \{0, 1\}$ is the human label, $\hat{Y} \in \{0, 1\}$ is the LLM-judged label, and $R \in \{0, 1\}$ indicates calibration ($R = 1$, Y observed) versus test ($R = 0$, Y missing). We assume observations are i.i.d. and $R \perp (Y, \hat{Y})$ with labeled ratio $\Pr(R = 1) = \frac{m}{m+n}$.

A useful way to motivate the EIF here is to note two sources of information about $\theta = \mathbb{E}[Y]$: (i) the large test sample estimates the distribution of $\hat{Y}$ precisely; and (ii) the calibration sample links Y to $\hat{Y}$. The regression $\mu(\hat{y}) = \mathbb{E}[Y | \hat{Y} = \hat{y}]$, $\hat{y} \in \{0, 1\}$, is the optimal way to translate $\hat{Y}$ into Y : by iterated expectations, $\theta = \mathbb{E}[\mathbb{E}(Y | \hat{Y})] = \mathbb{E}[\mu(\hat{Y})]$. This suggests an impute-then-average estimator that averages $\mu(\hat{Y})$ over the test set; since μ is unknown, we estimate it from the calibration sample.

Proposition 3.1 (Efficient influence function). *Given $(Y, \hat{Y}, R) \sim P$ with R independent of Y and $\hat{Y}$, the efficient influence function for $\theta = \mathbb{E}[Y]$ under P is*

$$\phi^*(R, Y, \hat{Y}; P) = \mu(\hat{Y}) - \theta + \frac{R}{\Pr(R=1)} \{Y - \mu(\hat{Y})\}. \quad (12)$$

Proposition 3.2. *Define $\hat{\mu}(1) := \frac{\sum_{j=1}^m \mathbf{1}\{\hat{Y}_j=1\}Y_j}{\sum_{j=1}^m \mathbf{1}\{\hat{Y}_j=1\}}$, $\hat{\mu}(0) := \frac{\sum_{j=1}^m \mathbf{1}\{\hat{Y}_j=0\}Y_j}{\sum_{j=1}^m \mathbf{1}\{\hat{Y}_j=0\}}$, and $\hat{\mu}(\hat{Y}) = \hat{\mu}(0)\mathbf{1}\{\hat{Y} = 0\} + \hat{\mu}(1)\mathbf{1}\{\hat{Y} = 1\}$. Then an asymptotically efficient estimator for θ under (1) is*

$$\hat{\theta}_{\text{EIF}} = \frac{1}{N} \sum_{i=1}^N \hat{\mu}(\hat{Y}_i) + \frac{1}{m} \sum_{j=1}^m (Y_j - \hat{\mu}(\hat{Y}_j)), \quad (13)$$

and $\sqrt{N}(\hat{\theta}_{\text{EIF}} - \theta) \xrightarrow{d} \mathcal{N}(0, V_{\text{EIF}})$, where $a = q_0 + q_1 - 1$, $v_0 = q_0(1 - q_0)$, $v_1 = q_1(1 - q_1)$, and

$$V_{\text{EIF}} = \frac{\theta(1-\theta)}{p(1-p)} [\theta(1-\theta)a^2 + (\gamma_1 + 1)(\theta v_1 + (1-\theta)v_0)]. \quad (14)$$

PPI is not generally efficient. Efficiency theory implies that the EIF estimator $\hat{\theta}_{\text{EIF}}$ in (13) is asymptotically efficient under the general i.i.d. model for $(Y, \hat{Y}, R) \sim P$, and thus has the smallest asymptotic variance among all regular, asymptotically linear estimators. With some algebra, one can verify that all estimators in Section 2 are regular and asymptotically linear. In particular, PPI corresponds to the special choice $\mu(\hat{Y}) = \hat{Y}$ in (12). This is not generally efficient unless $\mathbb{E}[Y | \hat{Y}] = \hat{Y}$, i.e., $\hat{Y}$ perfectly predicts Y .

Optimally tuned PPI++ matches the EIF in the binary case. Unlike PPI, PPI++ searches over a one-parameter family of unbiased estimators and chooses λ to minimize asymptotic variance. In general it need not equal the EIF, since $\mathbb{E}[Y | \hat{Y}]$ need not be of the form $\lambda \hat{Y}$. However, when $Y, \hat{Y} \in \{0, 1\}$, the regression $\mu(\hat{Y}) = \mathbb{E}[Y | \hat{Y}]$ is always linear in $\hat{Y}$, so the PPI++ family is rich enough to match the EIF. Letting $\gamma_1 = n/m$, Proposition 2.3 gives the variance-minimizing choice $\lambda^* = \frac{\gamma_1 \text{Cov}(Y, \hat{Y})}{(1 + \gamma_1) \text{Var}(\hat{Y})} = \frac{\gamma_1 \theta(1-\theta)(q_0 + q_1 - 1)}{(1 + \gamma_1)p(1-p)}$, where $p = (1 - \theta)(1 - q_0) + \theta q_1$. By Bayes' rule, $\mu(1) = \Pr(Y = 1 | \hat{Y} = 1) = \frac{\theta q_1}{p}$, $\mu(0) = \Pr(Y = 1 | \hat{Y} = 0) = \frac{\theta(1-q_1)}{1-p}$. A short calculation

shows $\mu(1) - \mu(0) = \frac{\theta(1-\theta)(q_0+q_1-1)}{p(1-p)}$, and hence $\lambda^* = \frac{\gamma_1}{1+\gamma_1} \{\mu(1) - \mu(0)\}$, i.e., the slope of the linear form of $\mu(\hat{Y})$. Therefore, optimally tuned PPI++ is asymptotically equivalent to the EIF estimator in the binary setting.

MLE is equivalent to EIF Under the correctly specified model $(Y, \hat{Y}, R) \sim P$, classical likelihood theory implies that the MLE is asymptotically efficient for θ (Proposition 2.4) and therefore attains the smallest possible asymptotic variance (Bickel et al., 1993). In the binary-outcome case, the model is saturated by the three parameters (θ, q_0, q_1) , and thus we expect the asymptotic variance of the EIF estimator to coincide with that of the MLE. The detailed algebra establishing the equivalence of the variance expressions is deferred to the appendix.

RG is less efficient than PPI. The following proposition shows that PPI is more efficient than Rogan–Gladden in the binary misclassification model.

Proposition 3.3. *Fix $\theta \in (0, 1)$ and $q_0, q_1 \in (0, 1)$ with $q_0 + q_1 - 1 > 0$, and assume $n, m \rightarrow \infty$ with $n/m \rightarrow \gamma_1 \in (0, \infty)$. Then*

$$V_{\text{PPI}} \leq V_{\text{RG}},$$

where V_{RG} and V_{PPI} are defined in (6) and (7), respectively. Equality holds if and only if $q_0 = q_1 = 1$ (equivalently, $\Pr(\hat{Y} = Y) = 1$).

Asymptotic Efficiency Summary. The relationships derived above imply a clear ordering of performance. Under the general i.i.d. model for $(Y, \hat{Y})$, the EIF estimator $\hat{\theta}_{\text{EIF}}$ attains the semiparametric efficiency bound. In the binary case, optimally tuned PPI++ is asymptotically equivalent to $\hat{\theta}_{\text{EIF}}$, and consequently to the MLE. Formally, for $N = n + m \rightarrow \infty$ and $\frac{n}{m} \rightarrow \gamma_1$:

$$\begin{aligned} \hat{\theta}_{\text{EIF}} &=_{\text{asympt}} \hat{\theta}_{\text{MLE}} =_{\text{asympt}} \hat{\theta}_{\text{PPI++}}(\lambda^*) \\ &\prec_{\text{asympt}} \hat{\theta}_{\text{PPI}} \prec_{\text{asympt}} \hat{\theta}_{\text{RG}}. \end{aligned}$$

where $=_{\text{asympt}}$ denotes equivalence up to $o_p(n^{-1/2})$ and $\prec_{\text{asympt}}$ denotes smaller asymptotic variance.

4. Beyond Binary Outcomes

In the binary case, $\mu(\hat{Y}) = \mathbb{E}[Y | \hat{Y}]$ is necessarily linear, allowing optimally tuned PPI++ to achieve the efficiency bound. Moreover, the nonparametric distribution on $(Y, \hat{Y})$ is equivalent to the three-parameter distribution used for the MLE in the binary case; this “saturated” property allows the MLE to attain the efficiency bound *without* making any restrictive parametric assumptions.

For richer surrogate outputs $\hat{Y}$ (e.g., continuous scores and ordinal ratings), the EIF—and thus $\hat{\theta}_{\text{EIF}}$ in (13)—is unchanged. However, $\mu(\cdot)$ may be nonlinear, and efficiency depends on how close $\hat{\mu}(\cdot)$ is to the true conditional mean.

When both Y and $\hat{Y}$ are one-dimensional, estimating $\mu(\hat{Y})$ reduces to a univariate regression problem, which can be addressed using models ranging from simple linear regression to flexible nonparametric smoothers. Given a consistent estimator $\hat{\mu}(\cdot)$, we can then construct the plug-in EIF estimator:

$$\hat{\theta}_{\text{EIF}} = \frac{1}{N} \sum_{i=1}^N \left(\hat{\mu}(\hat{Y}_i) + \frac{1}{\hat{P}(R=1)} R_i \{Y_i - \hat{\mu}(\hat{Y}_i)\} \right). \quad (15)$$

Under the same MCAR random-labeling design used throughout the paper, (15) remains asymptotically efficient as long as $\hat{\mu}(\cdot)$ consistently estimates $\mu(\cdot)$. If one instead adaptively designs the labeling process R , the semiparametric model changes and the corresponding efficient correction must be modified; we view that as a natural extension rather than part of the present theorem package.

The same principle extends beyond simple random labeling when the labeling rule is observed. If labels are sampled under a missing-at-random mechanism with known or consistently estimable probability $\pi(W) = \Pr(R=1 | W)$ for observed information W (e.g., $W = X$, $W = \hat{Y}$, or $W = (X, \hat{Y})$), the efficient correction replaces $\mu(\hat{Y})$ by $\mu(W) = \mathbb{E}[Y | W]$ and $P(R=1)$ by $\pi(W)$:

$$\phi_W^*(O) = \mu(W) - \theta + \frac{R}{\pi(W)} \{Y - \mu(W)\}.$$

Thus covariate-dependent labeling and context-dependent judge reliability can be handled by conditioning and inverse-probability weighting. By contrast, when the decision to label depends on unobserved Y or latent difficulty, the mean is not identifiable without additional assumptions; in such cases a stable misclassification model may make RG-style correction more transportable than EIF calibration.

Finally, unlike the binary (or the categorical with finite categories) case, we do not have a direct analogy with the MLE or RG estimator in the general case without assuming additional parametric relationship of Y and $\hat{Y}$.

5. Simulation Study

5.1. Binary Y

We compare the performance of following estimators in simulation: the naive estimator in (3), the Rogan–Gladden estimator in (6), the PPI estimator in (7), the PPI++ estimator in (8), the MLE in (10), and the EIF estimator in (13).

5.2. Simulation Setup

Data-generating process. Each replicate consists of $N = 2,000$ i.i.d. samples. The true binary label is generated as $Y_i \stackrel{\text{i.i.d.}}{\sim} \text{Bern}(\theta)$, with $\theta \in \{0.1, 0.2, \dots, 0.9\}$.

The LLM-as-a-judge surrogate $\hat{Y}_i$ is generated from a misclassification model with sensitivity $\Pr(\hat{Y}_i = 1 | Y_i = 1) = q_1$ and specificity $\Pr(\hat{Y}_i = 0 | Y_i = 0) = q_0$.

Parameter grid. We vary the following parameters across a full factorial design:

- True prevalence: $\theta \in \{0.1, 0.2, \dots, 0.9\}$.
- Judge accuracy: $q_0, q_1 \in \{0.6, 0.7, 0.8\}$.
- Labeling budget (calibration size): $\{1\%, 5\%, 10\%$ of N (i.e., $m \in \{20, 100, 200\}$ in expectation).

The calibration set is obtained by *random sampling* from the full dataset. For each configuration, we perform $B = 1,000$ Monte Carlo replicates to compute bias as well as coverage and average width of 90% confidence intervals.

Confidence interval construction. We apply a logit transformation to all confidence intervals to improve finite-sample coverage when θ is near the boundary (Brown et al., 2001; Stone, 1996). Given point estimate $\hat{\theta}$ and variance estimate $\hat{V}$, we construct the $1 - \alpha$ confidence interval as $\text{CI}_{1-\alpha} = \text{expit}\left(\text{logit}(\hat{\theta}) \pm z_{\alpha/2} \cdot \frac{\sqrt{\hat{V}}}{\hat{\theta}(1-\hat{\theta})}\right)$, where the standard error on the logit scale is obtained via the delta method (Van der Vaart, 2000). As a robustness check requested by the reviewers, we also reran the binary simulation with standard truncated Wald intervals in place of the logit transform; the qualitative ranking was unchanged. Appendix A.2 reports the comparison in table form.

Evaluation metrics. For each estimator $\hat{\theta}$, let $\hat{\theta}^{(b)}$ denote its value in replicate $b = 1, \dots, B$, and let $\text{CI}^{(b)} = [L^{(b)}, U^{(b)}]$ denote its corresponding 90% confidence interval. We compute:

- **Bias:** $\text{Bias} = \frac{1}{B} \sum_{b=1}^B (\hat{\theta}^{(b)} - \theta)$;
- **Coverage probability:** $\text{Coverage} = \frac{1}{B} \sum_{b=1}^B \mathbf{1}\{L^{(b)} \leq \theta \leq U^{(b)}\}$;
- **Mean confidence interval width:** $\text{Width} = \frac{1}{B} \sum_{b=1}^B (U^{(b)} - L^{(b)})$.

5.3. Results Under Symmetric Judge Accuracy ($q_0 = q_1$)

Figures 1–3 are organized with rows corresponding to accuracy levels $q_0 = q_1 \in \{0.6, 0.7, 0.8\}$ and columns corresponding to labeling budgets.

Bias. Figure 1 shows that all estimators except the naive one exhibit smaller average bias than the naive sample-mean estimator. Recall that the bias of the naive estimator is $(1 - q_0) + \theta(q_0 + q_1 - 2)$, which is negative for large $\theta > 0.5$ and positive for small $\theta < 0.5$, consistent with our simulation results. Among the remaining estimators, $\hat{\theta}_{\text{RG}}$ exhibits comparatively larger finite-sample bias.

Coverage. Figure 2 reports empirical coverage rates. The naive estimator exhibits severe undercoverage, consistent with its asymptotic invalidity (at $\theta = 0.5$ the naive estimator is unbiased so the coverage is closer to nominal level). All other estimators maintain near-nominal coverage across

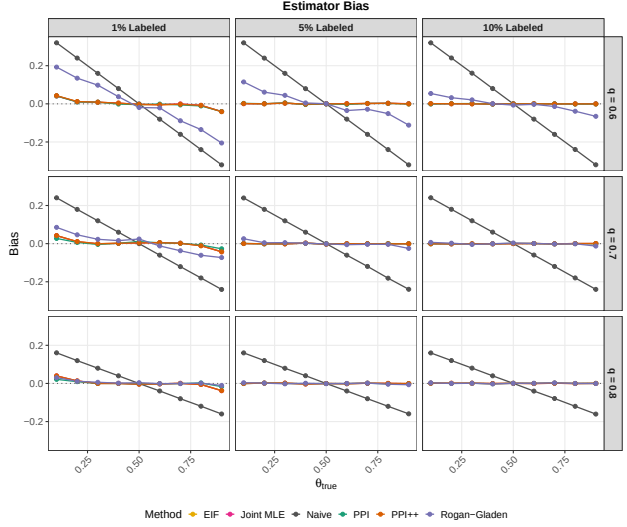


Figure 1. Estimator bias of $\hat{\theta}$. All except for naive estimators achieve near-zero bias; the naive estimator exhibits large bias in many settings. When only 1% of the data is labeled, $\hat{\theta}_{\text{RG}}$ also exhibits considerable bias.

settings. The RG estimator and its associated confidence intervals tend to overcover, particularly when we have limited number of labeled observations in the calibration set.

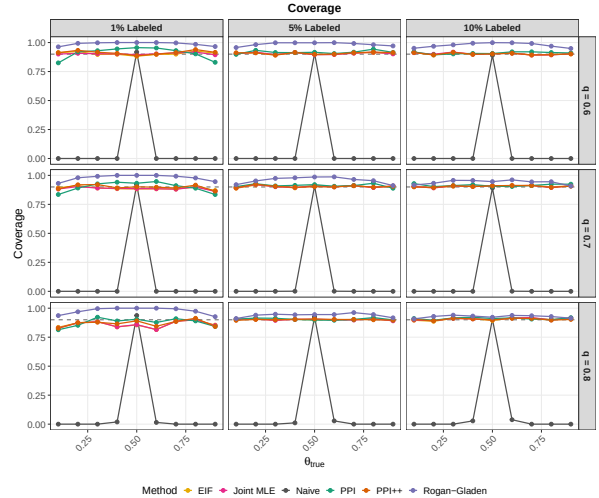


Figure 2. Empirical coverage of estimators. The naive estimator undercovers, while RG, PPI, and PPI++ are near-nominal; RG often overcovers with few labels and weak judges.

Confidence interval width. Figure 3 compares interval widths among valid estimators (i.e., $\hat{\theta}_{\text{naive}}$ excluded). As our theory predicted in Section 3, EIF and PPI++ produce nearly identical and consistently shortest intervals, typically 35–55% narrower than standard PPI. MLE intervals are slightly wider but overall comparable to EIF and PPI++. Rogan–Gladen intervals widen dramatically under low-accuracy judges due to the $1/(q_0 + q_1 - 1)^2$ amplification factor in its asymptotic variance—at $q_0 = q_1 = 0.6$, RG intervals are roughly 10× wider than those of EIF/PPI++.

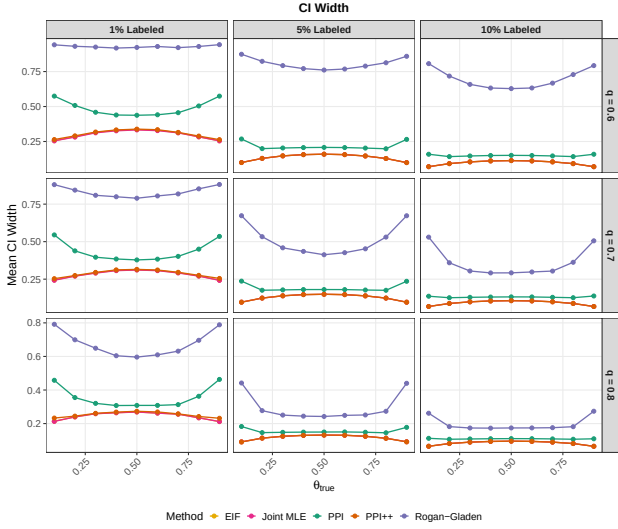


Figure 3. Average CI width in simulated data. EIF and PPI++ are shortest and nearly identical, outperforming Rogan-Gladen by 3–15 $\times$, with the largest gains when $q_0 + q_1 - 1$ is small. Naive CIs are omitted for undercoverage.

5.4. Beyond Random Labeling

We also stress-tested the methods when the calibration labels are not an i.i.d. random subset. Table 1 summarizes binary simulations with $N = 5,000$, 10% labels, and 1,000 replicates. When the sampling rule depends only on observed quantities, EIF remains valid after replacing $P(R = 1)$ with the correct $\Pr(R = 1 \mid X, \hat{Y})$; RG can fail sharply when the selected calibration set no longer represents the confusion matrix. In a separate context-dependent simulation with $X \in \{0, 1\}$ indexing easy/hard examples, using $\mathbb{E}[Y \mid X, \hat{Y}]$ rather than $\mathbb{E}[Y \mid \hat{Y}]$ preserved nominal coverage and reduced mean CI width from 0.104 to 0.089. Appendix A.4 gives the full mechanism table and the MNAR comparisons.

Sampling Rule	EIF Cov.	RG Cov.	EIF / RG Width
MCAR	0.937	0.963	0.046 / 0.062
MAR(X)	0.912	0.953	0.047 / 0.062
MAR($\hat{Y}$)	0.972	0.000	0.052 / 0.106
MAR($X, \hat{Y}$)	0.951	0.088	0.050 / 0.075

Table 1. Non-MCAR labeling simulations. EIF uses the correct observed sampling probability $\Pr(R = 1 \mid X, \hat{Y})$ when needed.

5.5. Continuous Y

Here, we consider the simulation setup in Ji et al. (2025) where $\hat{Y}$ takes discrete values (e.g., LLM ratings on a 1–5 scale) while the true outcome Y is continuous. This scenario is common when LLMs are used as judges, providing ordinal ratings, but the ultimate goal is to infer the continuous quality scale for the models Y .

Data Generating Process. Let $Z \sim \text{Unif}(\{1, 2, 3\})$ denote

a latent mixture component, with conditional distribution $Y \mid Z \sim \mathcal{N}(\mu_Z, \sigma^2)$ where $\mu = (\mu_1, \mu_2, \mu_3)^\top$. The surrogate prediction is on the component Z , and it’s easy to see that we need to carefully model $\mathbb{E}[Y \mid \hat{Y}]$. Even when the prediction $\hat{Y}$ perfectly identifies the latent class, the naive estimator in (3) is biased for $\mathbb{E}[Y]$ because $\hat{Y}$ and Y are on different scales. Specifically, the bias would be $2 - (\mu_1 + \mu_2 + \mu_3)/3$, which grows as the μ_k values diverge from their category indices.

Simulation Design. We fix $N = 2000$, $\mu_1 = 1$, $\mu_2 = 2$, $\sigma = 1$, and vary $\mu_3 \in \{3, 4, \dots, 9\}$ to control bias magnitude. The labeled fraction varies over $\{5\%, 10\%, 20\%\}$. We report coverage of 90% confidence intervals and RMSE across $B = 500$ replications.

We compare the naive estimator (3), the prediction-powered estimator (7), the tuned PPI++ estimator (8), and the EIF estimator (15). For the EIF, we consider three ways to approximate $\mathbb{E}[Y \mid \hat{Y}]$: (i) EIF, which estimates $g(z) = \mathbb{E}[Y \mid \hat{Y} = z]$ separately for $z \in \{1, 2, 3\}$; (ii) EIF-LINEAR, which fits $g(\hat{Y}) = a + b\hat{Y}$ via linear regression (asymptotically equivalent to PPI++); and (iii) EIF-GAM/SPLINE, which uses nonparametric calibration with $\hat{Y}$ treated as a smooth covariate. Since we impose no additional distributional assumptions on Y , parametric misclassification models such as Rogan-Gladen or MLE are not applicable.

Results. Figure 4 shows how inferential performance depends on estimating $\mathbb{E}[Y \mid \hat{Y}]$. When $\mu = (1, 2, 3)$, the conditional mean is linear in $\hat{Y}$, so $\mathbb{E}[Y \mid \hat{Y}] = \hat{Y}$ is correctly specified and all methods except the naive estimator achieve nominal coverage. Confidence-interval widths are then indistinguishable, since EIF, PPI, and PPI++ are asymptotically equivalent. As μ_3 increases, the conditional mean becomes nonlinear. Although all debiased estimators retain valid coverage, a clear efficiency ordering emerges: PPI is least efficient, followed by PPI++ and EIF with a linear model for $\mathbb{E}[Y \mid \hat{Y}]$ (which are asymptotically equivalent under optimal tuning). Using flexible models for $\mathbb{E}[Y \mid \hat{Y}]$, such as GAMs (Wood, 2012), splines, or a three-category model estimating $g(z) = \mathbb{E}[Y \mid \hat{Y} = z]$ for $z \in \{1, 2, 3\}$, yields the most efficient performance.

6. Real Data Application

6.1. Data and Experimental Setup

We evaluate the estimators on real human preference data from the arena-human-preference-140k dataset, which is collected on Chatbot Arena (Chiang et al., 2024), an online platform where users compare responses from two AI models and select a winner. We focus on three high-frequency model pairs, all with Claude Opus 4 as model A: (1) Claude Opus 4 vs. Gemini 2.5 Flash ($n = 493$); (2)

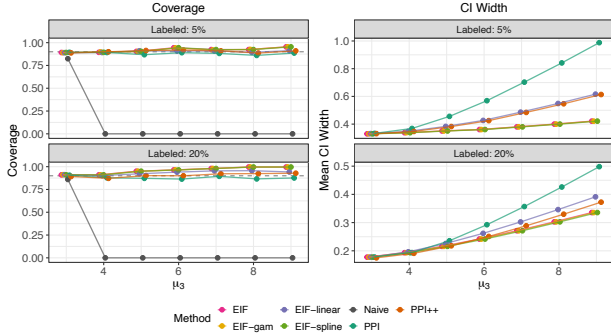


Figure 4. Coverage (left; dashed = 90%) and CI width (right) in discrete-prediction simulations versus bias μ_3 . Naive CIs under-cover as bias grows; others remain near-nominal. EIF with spline/GAM calibration is narrowest; linear correction $\approx$ PPI++.

Claude Opus 4 vs. Gemini 2.5 Pro ($n = 414$); (3) Claude Opus 4 vs. Qwen3-235B ($n = 494$).

For each battle, the ground truth $Y_i = 1$ if the human selected model A (Claude), and $Y_i = 0$ otherwise. We deploy two LLM judges, GPT-4o-mini and GPT-5.2, to produce surrogate labels $\hat{Y}_i \in \{0, 1\}$ based on the same prompt-response pairs shown to human raters. In additional experiments run after the submission, we also evaluated Claude Sonnet 4.6 and Gemini 2.5 Pro. The same qualitative ordering persisted, with EIF / PPI++ / MLE near nominal coverage and substantially tighter intervals than Rogan–Gladden; Appendix A.3 reports the exact aggregated coverage/width table for these added judge families.

To mimic a real-world setting where we have only access to part of the preference data, we use a 10% random sample of the data as a labeled calibration set to estimate either $\mathbb{E}[Y | \hat{Y}]$ or the misclassification probabilities (q_0, q_1) , and the remaining 90% as an unlabeled test set. All confidence intervals are constructed at the 90% nominal level.

6.2. Results

Judge quality. Table 2 reports estimated specificity ($\hat{q}_0$) and sensitivity ($\hat{q}_1$) for each judge–dataset pair, computed from a randomly sampled 10% calibration set. Values with $q_0 + q_1 > 1$ indicate discriminative performance beyond random guessing.

Judge quality varies substantially across settings. GPT-4o-mini performs best on Claude 4 vs. Gemini 2.5 Flash, but degrades on Claude 4 vs. Gemini 2.5 Pro. GPT-5.2 achieves the strongest overall discrimination on Claude 4 vs. Gemini 2.5 Pro. Both judges perform poorly on Claude 4 vs. Qwen3-235B, exhibiting low specificity despite moderate sensitivity. These heterogeneous error profiles underscore the need for bias correction: naive estimators directly inherit judge-specific systematic errors.

Coverage and efficiency. Figure 5 summarizes empirical

Model Pair	Judge	$\hat{q}_0$	$\hat{q}_1$
Claude vs. Gemini Flash	GPT-4o-mini	0.74	0.69
Claude vs. Gemini Flash	GPT-5.2	0.47	0.67
Claude vs. Gemini Pro	GPT-4o-mini	0.50	0.81
Claude vs. Gemini Pro	GPT-5.2	0.64	0.82
Claude vs. Qwen3	GPT-4o-mini	0.44	0.70
Claude vs. Qwen3	GPT-5.2	0.50	0.64

Table 2. Estimated judge quality metrics from calibration data ($\hat{q}_0$: specificity; $\hat{q}_1$: sensitivity).

coverage versus mean CI width across 1,000 random calibration/test splits. The naive estimator is unreliable: its coverage varies sharply with the magnitude and direction of judge bias, achieving near-perfect coverage only when systematic errors happen to align with the true prevalence (e.g., GPT-5.2 on Gemini 2.5 Pro), and collapsing to 0% coverage when bias is substantial (e.g., GPT-4o-mini on Gemini 2.5 Pro and both judges on Qwen3-235B).

In contrast, PPI++, EIF, and MLE achieve near-nominal coverage across all judge–dataset combinations, with mean interval widths between 0.27 and 0.30. PPI slightly overcovers with wider intervals (0.35–0.44), while Rogan–Gladden is highly conservative, attaining 100% coverage at the cost of very wide intervals (0.76–0.96). Among methods with valid coverage, PPI++, EIF, and MLE provide the best coverage–width tradeoff, achieving nominal coverage with intervals roughly three times narrower than Rogan–Gladden, consistent with our theoretical results.

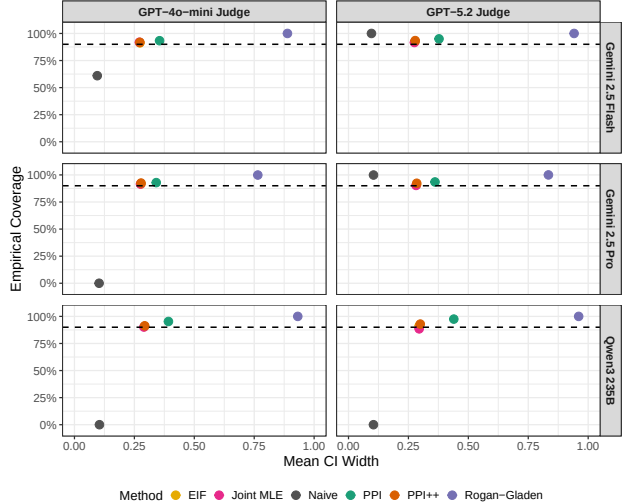


Figure 5. Empirical coverage versus mean CI width across 1,000 random 90/10 test/calibration splits. Dashed line indicates 90% nominal coverage. Error bars show 90% confidence intervals. Naive estimation (gray) achieves 0% coverage when judge bias is substantial, while PPI++, EIF, and MLE maintain nominal coverage with narrow intervals.

An illustrative example. To illustrate the coverage–width tradeoff in Figure 5, we display point estimates and 90% confidence intervals for one random calibration/test split (Claude 4 vs. Qwen3) in Figure 6. The horizontal black line denotes the true win rate from all human labels. We observe

that: (i) the naive estimator is biased, overestimating the win rate with GPT-4o-mini; (ii) Rogan–Gladden produces extremely wide, largely uninformative intervals; and (iii) PPI++, EIF, and MLE yield similar point estimates with much narrower intervals that cover the truth.

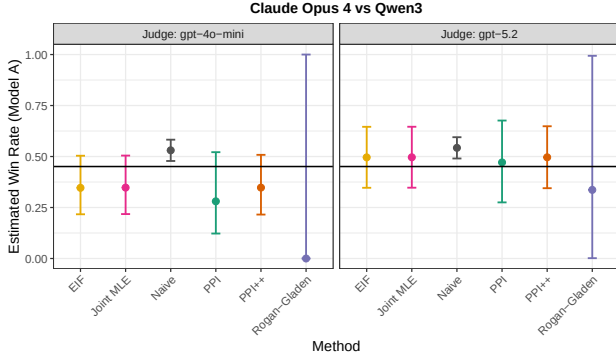


Figure 6. Point estimates and 90% confidence intervals for win rate estimation across three model pairs, using two LLM judges (GPT-4o-mini and GPT-5.2). Horizontal line is the true win rate.

7. Discussion

We presented a unified view of calibration in LLM-as-a-judge evaluations via efficient influence functions. We analyzed, theoretically and empirically, two complementary approaches: measurement-error correction via misclassification models and surrogate-based methods such as PPI. For mean parameters—the common target in LLM-as-a-judge settings—both Rogan–Gladden and PPI-type estimators yield simple, reliable corrections, with PPI++/EIF-based methods preferred for their efficiency gains. In the binary case, optimally tuned PPI++ coincides with the EIF estimator. We emphasize, however, that these efficiency conclusions are strongest under the same-distribution calibration regime studied here; under distribution shift, RG-style correction can be more transportable when a stable misclassification model is more plausible than a stable full calibration relation.

Several extensions follow. A key direction is instance-dependent misclassification, where error rates vary with features or difficulty. Extending our framework parallels semiparametric efficiency theory by replacing $\mu(\hat{Y})$ with $\mu(X, \hat{Y})$ and the constant labeling probability with $\Pr(R = 1 | X, \hat{Y})$. Another direction is multi-judge ensembling to improve robustness, which requires modeling distinct judge error rates (q_0, q_1) in (4) and principled aggregation of surrogate labels (e.g., optimal weighting or learned combinations). Finally, robustness to distribution shift deserves further study, including covariate shift (e.g., response-style differences between calibration and test sets) and label shift (e.g., changes in $\Pr(Y = 1)$ across deployment contexts).

As LLM-as-a-judge evaluations become central to model

development and scientific assessment, our work provides a practical foundation for valid and efficient use of limited human labels.

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Impact Statement

This paper advances methods for statistically valid evaluation when LLM outputs are used as surrogate labels. The work is intended to improve the reliability and transparency of LLM-as-a-judge pipelines, and we emphasize that human oversight remains necessary in high-stakes settings.

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Appendix for Efficient Inference for Noisy LLM-as-a-Judge Evaluation

A. Additional Results

A.1. Finite-Sample Variability in Calibration Estimates

The Rogan–Gladen estimator relies on plug-in estimates $\hat{q}_0$ and $\hat{q}_1$. Figure 7 illustrates their finite-sample variability across simulation settings. While these estimates are approximately unbiased, their variance can be substantial when the labeling budget is small. At 1% labeling ($m = 20$), the positive class may contain only $m_1 \approx 3$ observations, yielding an RMSE of 0.3 for $\hat{q}_1$, compared to 0.1 for $\hat{q}_0$.

Because the RG estimator divides by $\hat{q}_0 + \hat{q}_1 - 1$, the resulting confidence intervals can be wide when m is small or when class imbalance leaves m_1 (or m_0) with few observations.

In contrast, PPI-type estimators and EIF-based approaches do not require explicit estimation of (q_0, q_1) . Instead, they exploit unbiased correction to learn $\mathbb{E}[Y \mid \hat{Y}]$, making them more stable in small-calibration regimes.

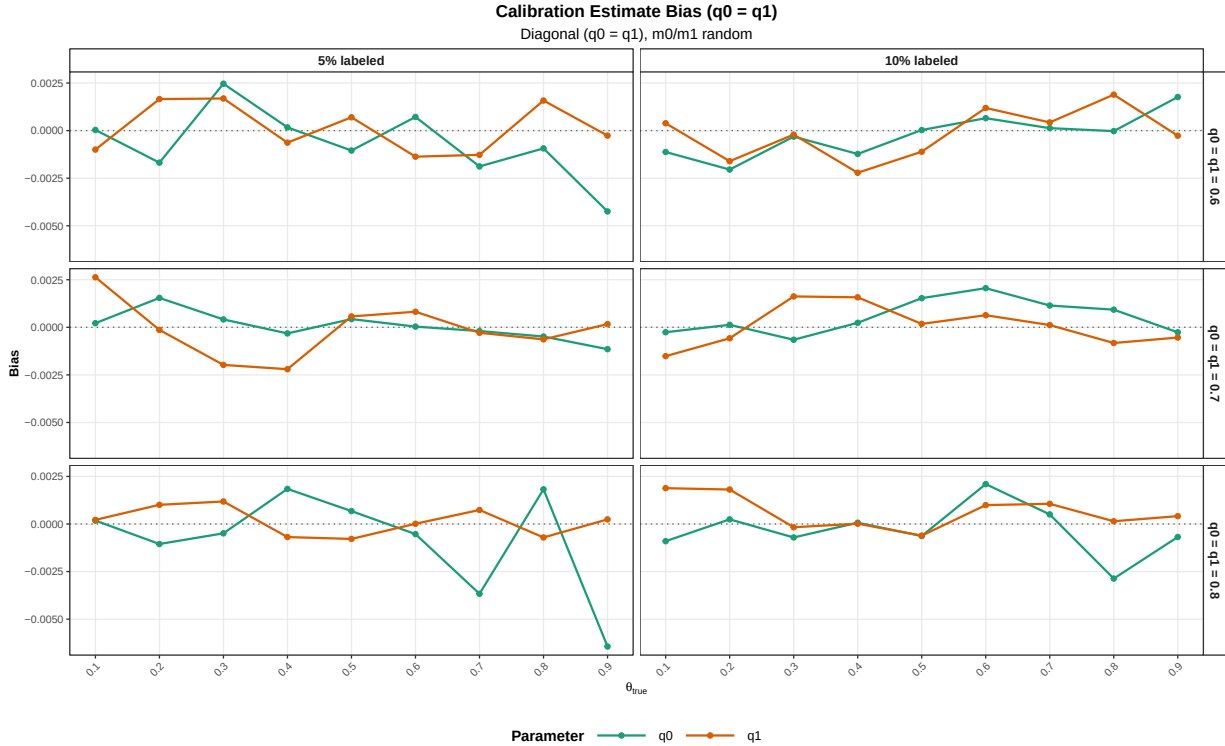


Figure 7. Finite-sample RMSE of calibration estimates $\hat{q}_0$ and $\hat{q}_1$.

A.2. Binary CI Robustness: Logit vs. Truncated Wald

Table 3 summarizes a robustness check for the binary-mean simulation in which we replace the logit-transformed confidence intervals in the main text with standard asymptotic Wald intervals truncated to $[0, 1]$. The qualitative ranking is unchanged. We include Rogan–Gladen and Naive explicitly as well, so the comparison is complete for all methods rather than only for the methods whose CI construction changed.

A.3. Additional LLM Judges Beyond GPT

To check that the real-data conclusions are not specific to GPT-family judges, we expanded the Arena-style experiment to four judges by adding Claude Sonnet 4.6 and Gemini 2.5 Pro. Table 4 reports average empirical coverage and mean interval width over 1,000 random 90/10 calibration/test splits, aggregated over the three datasets in Section 6. This table is the numerical companion to the real-data coverage/width plot, and it makes clear that the same qualitative pattern persists across judge families: EIF / PPI++ / MLE remain near nominal coverage with substantially narrower intervals than Rogan–Gladen,

Efficient Inference for Noisy LLM-as-a-Judge Evaluation

Method	Cov. (logit)	Cov. (wald)	Width (logit)	Width (wald)
EIF	0.899	0.895	0.0696	0.0696
Joint MLE	0.899	0.896	0.0696	0.0696
PPI++	0.899	0.895	0.0695	0.0695
PPI	0.907	0.899	0.0960	0.0950
Rogan–Gladen	0.920	0.920	0.2388	0.2388
Naive	0.061	0.061	0.0234	0.0234

Table 3. Average performance across all 162 binary-mean simulation settings under logit-transformed versus truncated-Wald confidence intervals. The main finite-sample sensitivity is concentrated in PPI near the boundary; EIF / PPI++ / Joint MLE change only minimally, and the overall method ordering is unchanged.

while the naive estimator remains highly sensitive to judge-specific bias.

Method	Claude Sonnet 4.6	Gemini 2.5 Pro	GPT-4o-mini	GPT-5.2
EIF	0.913 / 0.274	0.910 / 0.276	0.909 / 0.281	0.913 / 0.285
PPI++	0.916 / 0.274	0.915 / 0.276	0.918 / 0.282	0.918 / 0.286
Joint MLE	0.912 / 0.272	0.914 / 0.274	0.912 / 0.279	0.909 / 0.284
PPI	0.938 / 0.345	0.931 / 0.345	0.942 / 0.364	0.959 / 0.391
Rogan–Gladen	0.976 / 0.778	0.975 / 0.808	0.981 / 0.857	0.994 / 0.904
Naive	0.675 / 0.102	0.000 / 0.098	0.207 / 0.101	0.666 / 0.101

Table 4. Average empirical coverage / mean CI width for the added four-judge real-data experiment. Each entry aggregates 1,000 random splits over the three Arena-style datasets.

A.4. Labeling Mechanisms Beyond MCAR

Table 5 summarizes how common label-acquisition rules map to the missing-data taxonomy. The key distinction is whether the sampling probability can be written as a function of observed variables. If $\pi(W) = \Pr(R = 1 \mid W)$ is known or consistently estimable for observed W , the EIF estimator remains valid by using $\mu(W) = \mathbb{E}[Y \mid W]$ and inverse-probability weighting. If labeling depends on unobserved Y or latent difficulty, neither $\pi(W)$ nor $\mu(W)$ identifies the population mean without stronger assumptions.

Labeling Rule	Type	LLM-Judge Example	Main Implication
Random subset	MCAR	Simple random annotation	EIF and RG valid
Based on X	MAR(X)	Select by topic or length	EIF valid with $\pi(X)$
Based on $\hat{Y}$	MAR($\hat{Y}$)	Label by judge confidence	EIF valid with $\pi(\hat{Y})$; RG may fail
Based on $X, \hat{Y}$	MAR($X, \hat{Y}$)	Stratified active auditing	EIF valid with $\pi(X, \hat{Y})$
Based on latent difficulty	MNAR	Label unusually hard cases	Not identifiable without assumptions

Table 5. Labeling mechanisms and estimator validity. RG-style correction can remain useful under some shifts if the misclassification mechanism is stable, while EIF calibration is most reliable when the labeling rule is observed.

Table 6 reports the MAR simulations summarized in the main text. The failure modes are complementary: under MAR($\hat{Y}$), RG estimates the confusion matrix on a selected sample and can collapse, whereas EIF remains valid after the sampling-probability adjustment.

Scenario	EIF Coverage	EIF Width	RG Coverage	RG Width
MCAR	0.937	0.046	0.963	0.062
MAR(X)	0.912	0.047	0.953	0.062
MAR($\hat{Y}$)	0.972	0.052	0.000	0.106
MAR($X, \hat{Y}$)	0.951	0.050	0.088	0.075

Table 6. Binary non-MCAR simulations with $N = 5,000$, 10% labels, and 1,000 replicates. EIF uses the correct observed sampling probability in each MAR setting.

For MNAR-style shifts, no uniformly best correction exists. Table 7 shows cases where RG is more transportable because the confusion-matrix model is stable, and a case where RG fails because the judge itself changes.

A.5. Context-Dependent Judge Reliability

We simulated a binary context $X \sim \text{Bern}(0.5)$ indexing easy and hard examples. The prevalence was 0.8 for easy examples and 0.2 for hard examples; judge sensitivity/specificity were (0.95, 0.92) in the easy context and (0.55, 0.50) in the hard

Scenario	EIF Coverage	RG Coverage	What Changes
Hard-example labeling	0.000	0.935	Labeling depends on Y
Disagreement labeling	0.005	0.875	Label when $Y \neq \hat{Y}$
Judge updated	0.860	0.000	$P(\hat{Y} Y)$ shifts

 Table 7. MNAR and distribution-shift simulations with $N = 5,000$, 10% labels, and 200 replicates.

context. Table 8 shows that conditioning on X reduces residual variation and yields narrower intervals at comparable coverage.

Method	Coverage	CI Width	RMSE
EIF with $\mathbb{E}[Y \hat{Y}]$	0.926	0.104	0.028
EIF with $\mathbb{E}[Y X, \hat{Y}]$	0.920	0.089	0.025

 Table 8. Context-dependent simulation with $N = 2,000$, 10% calibration labels, and 500 replicates. Conditioning on X gives about 14% narrower CIs.

B. Proofs of Propositions

B.1. Proof of Proposition 2.1

Proof. Since the test sample consists of i.i.d. draws of $\hat{Y} \in \{0, 1\}$ with $p = \Pr(\hat{Y} = 1)$, it follows that $\hat{Y}_i \sim \text{Bern}(p)$ for $i = 1, \dots, n$. For the sample mean $\hat{p} = \frac{1}{n} \sum_{i=1}^n \hat{Y}_i$, as $n \rightarrow \infty$, by the central limit theorem, we have that

$$\sqrt{n}(\hat{p} - p) \xrightarrow{d} \mathcal{N}(0, p(1 - p)).$$

On the calibration sample, define

$$X_{ab} = \sum_{j=1}^m \mathbf{1}\{Y_j = a, \hat{Y}_j = b\}, \quad \pi_{ab} = \Pr(Y = a, \hat{Y} = b), \quad a, b \in \{0, 1\}.$$

Then $X = (X_{00}, X_{01}, X_{10}, X_{11})^\top$ satisfies

$$X \sim \text{Multinomial}(m; \pi_{00}, \pi_{01}, \pi_{10}, \pi_{11}),$$

where

$$\begin{aligned} \pi_{00} &= \Pr(Y = 0, \hat{Y} = 0) = q_0(1 - \theta), \\ \pi_{01} &= \Pr(Y = 0, \hat{Y} = 1) = (1 - q_0)(1 - \theta), \\ \pi_{10} &= \Pr(Y = 1, \hat{Y} = 0) = (1 - q_1)\theta, \\ \pi_{11} &= \Pr(Y = 1, \hat{Y} = 1) = q_1\theta. \end{aligned}$$

Let $\pi = (\pi_{00}, \pi_{01}, \pi_{10}, \pi_{11})^\top$ and $\hat{P}(R = 1)^{\text{cal}} = X/m$. As $m \rightarrow \infty$, by the multivariate central limit theorem,

$$\sqrt{m}(\hat{P}(R = 1)^{\text{cal}} - \pi) \xrightarrow{d} \mathcal{N}(0, \text{diag}(\pi) - \pi\pi^\top).$$

Since

$$\hat{q}_1 = \frac{\sum_{j=1}^m \mathbf{1}\{\hat{Y}_j = 1, Y_j = 1\}}{m_1}, \quad \hat{q}_0 = \frac{\sum_{j=1}^m \mathbf{1}\{\hat{Y}_j = 0, Y_j = 0\}}{m_0},$$

it follows that

$$\hat{q}_1 = \frac{X_{11}}{X_{11} + X_{10}} = \frac{\hat{\pi}_{11}^{\text{cal}}}{\hat{P}(R = 1)_{11}^{\text{cal}} + \hat{P}(R = 1)_{10}^{\text{cal}}}, \quad \hat{q}_0 = \frac{X_{00}}{X_{00} + X_{01}} = \frac{\hat{P}(R = 1)_{00}^{\text{cal}}}{\hat{P}(R = 1)_{00}^{\text{cal}} + \hat{P}(R = 1)_{01}^{\text{cal}}}.$$

Define $h : (0, 1)^4 \rightarrow (0, 1)^2$ by

$$h(x_{00}, x_{01}, x_{10}, x_{11}) = \left(\frac{x_{00}}{x_{00} + x_{01}}, \frac{x_{11}}{x_{10} + x_{11}} \right)^\top,$$

so that $h(\pi) = (q_0, q_1)^\top$ and $h(\hat{P}(R=1)^{\text{cal}}) = (\hat{q}_0, \hat{q}_1)^\top$. Since $\pi_{00} + \pi_{01} = 1 - \theta > 0$ and $\pi_{11} + \pi_{10} = \theta > 0$, the mapping h is continuously differentiable at π . Its Jacobian at π is

$$\nabla h(\pi) = \begin{pmatrix} \frac{\pi_{01}}{(\pi_{00} + \pi_{01})^2} & -\frac{\pi_{00}}{(\pi_{00} + \pi_{01})^2} & 0 & 0 \\ 0 & 0 & -\frac{\pi_{11}}{(\pi_{10} + \pi_{11})^2} & \frac{\pi_{10}}{(\pi_{10} + \pi_{11})^2} \end{pmatrix}.$$

As $m \rightarrow \infty$, by the multivariate delta method,

$$\sqrt{m} \begin{pmatrix} \hat{q}_0 - q_0 \\ \hat{q}_1 - q_1 \end{pmatrix} \xrightarrow{d} \mathcal{N}(0, \nabla h(\pi)(\text{diag}(\pi) - \pi\pi^\top)\nabla h(\pi)^\top).$$

We note that $\theta = \pi_{11} + \pi_{10}$; therefore a direct calculation yields

$$\nabla h(\pi)(\text{diag}(\pi) - \pi\pi^\top)\nabla h(\pi)^\top = \begin{pmatrix} \frac{q_0(1-q_0)}{1-\theta} & 0 \\ 0 & \frac{q_1(1-q_1)}{\theta} \end{pmatrix},$$

and

$$\sqrt{m} \begin{pmatrix} \hat{q}_0 - q_0 \\ \hat{q}_1 - q_1 \end{pmatrix} \xrightarrow{d} \mathcal{N}\left(0, \begin{pmatrix} \frac{q_0(1-q_0)}{1-\theta} & 0 \\ 0 & \frac{q_1(1-q_1)}{\theta} \end{pmatrix}\right).$$

Since $n/m \rightarrow \gamma_1 \in (0, \infty)$, we have $\sqrt{n} = \sqrt{n/m} \sqrt{m}$ and $\sqrt{n/m} \rightarrow \sqrt{\gamma_1}$. By Slutsky's theorem,

$$\sqrt{n} \begin{pmatrix} \hat{q}_0 - q_0 \\ \hat{q}_1 - q_1 \end{pmatrix} \xrightarrow{d} \mathcal{N}\left(0, \gamma_1 \begin{pmatrix} \frac{q_0(1-q_0)}{1-\theta} & 0 \\ 0 & \frac{q_1(1-q_1)}{\theta} \end{pmatrix}\right).$$

Moreover, since the test sample is independent of the calibration sample, $\hat{p}$ is independent of $(\hat{q}_0, \hat{q}_1)$. Therefore, as $n \rightarrow \infty$,

$$\sqrt{n} \begin{pmatrix} \hat{p} - p \\ \hat{q}_0 - q_0 \\ \hat{q}_1 - q_1 \end{pmatrix} \xrightarrow{d} \mathcal{N}\left(0, \text{diag}\left(p(1-p), \gamma_1 \frac{q_0(1-q_0)}{1-\theta}, \gamma_1 \frac{q_1(1-q_1)}{\theta}\right)\right).$$

Define $g(p, q_0, q_1) = \frac{p+q_0-1}{q_0+q_1-1}$, so that $\theta = g(p, q_0, q_1)$. Algebra yields that

$$\frac{\partial g}{\partial p} = \frac{1}{q_0 + q_1 - 1}, \quad \frac{\partial g}{\partial q_0} = \frac{1 - \theta}{q_0 + q_1 - 1}, \quad \frac{\partial g}{\partial q_1} = -\frac{\theta}{q_0 + q_1 - 1},$$

and hence

$$\nabla g(p, q_0, q_1) = \frac{1}{q_0 + q_1 - 1} (1, 1 - \theta, -\theta).$$

By the multivariate delta method,

$$\sqrt{n} (\hat{\theta}_{\text{RG}} - \theta) \xrightarrow{d} \mathcal{N}\left(0, \nabla g(p, q_0, q_1) \text{diag}\left(p(1-p), \gamma_1 \frac{q_0(1-q_0)}{1-\theta}, \gamma_1 \frac{q_1(1-q_1)}{\theta}\right) \nabla g(p, q_0, q_1)^\top\right).$$

With algebra and the assumption that $\frac{N}{n} = \frac{m+n}{n} \rightarrow \frac{1+\gamma_1}{\gamma_1}$, we have that

$$V_{\text{RG}} = \left(\frac{1+\gamma_1}{\gamma_1}\right) \frac{1}{(q_0 + q_1 - 1)^2} \{p(1-p) + \gamma_1 [(1-\theta)q_0(1-q_0) + \theta q_1(1-q_1)]\}.$$

□

B.2. Proof of Proposition 2.2

Proof. We first use the following decomposition

$$\hat{\mu} = \hat{p} = \frac{1}{n} \sum_{i=1}^n \hat{Y}_i, \quad \hat{\Delta} = \frac{1}{m} \sum_{j=1}^m (\hat{Y}_j - Y_j), \quad \hat{\theta}_{\text{PPI}} = \hat{\mu} - \hat{\Delta}.$$

Step 1: Moments of $\hat{\mu}$. Since $\hat{Y}_i \stackrel{\text{iid}}{\sim} \text{Bern}(p)$ in the test sample,

$$\mathbb{E}[\hat{\mu}] = \mathbb{E}[\hat{Y}] = p, \quad \text{Var}(\hat{\mu}) = \frac{1}{n} \text{Var}(\hat{Y}) = \frac{p(1-p)}{n}.$$

Step 2: Moments of $\hat{\Delta}$. In the calibration sample, the summands $\hat{Y}_j - Y_j$ are i.i.d., hence

$$\mathbb{E}[\hat{\Delta}] = \mathbb{E}[\hat{Y} - Y], \quad \text{Var}(\hat{\Delta}) = \frac{1}{m} \text{Var}(\hat{Y} - Y).$$

Because $\hat{Y}, Y \in \{0, 1\}$, we have $\hat{Y} - Y \in \{-1, 0, 1\}$ and

$$(\hat{Y} - Y)^2 = \mathbf{1}\{\hat{Y} \neq Y\}.$$

Therefore,

$$\mathbb{E}[(\hat{Y} - Y)^2] = \Pr(\hat{Y} \neq Y) = \Pr(\hat{Y} = 1, Y = 0) + \Pr(\hat{Y} = 0, Y = 1). \quad (16)$$

Using the definitions of q_0, q_1 ,

$$\Pr(\hat{Y} = 1, Y = 0) = (1 - q_0)(1 - \theta), \quad \Pr(\hat{Y} = 0, Y = 1) = (1 - q_1)\theta,$$

so (16) becomes

$$\mathbb{E}[(\hat{Y} - Y)^2] = (1 - \theta)(1 - q_0) + \theta(1 - q_1). \quad (17)$$

Next, since $\mathbb{E}[\hat{Y}] = p$ and $\mathbb{E}[Y] = \theta$,

$$\mathbb{E}[\hat{Y} - Y] = \mathbb{E}[\hat{Y}] - \mathbb{E}[Y] = p - \theta,$$

and hence

$$\text{Var}(\hat{Y} - Y) = \mathbb{E}[(\hat{Y} - Y)^2] - \{\mathbb{E}[\hat{Y} - Y]\}^2 = (1 - \theta)(1 - q_0) + \theta(1 - q_1) - (p - \theta)^2. \quad (18)$$

Combining with $\text{Var}(\hat{\Delta}) = m^{-1} \text{Var}(\hat{Y} - Y)$ yields

$$\text{Var}(\hat{\Delta}) = \frac{(1 - \theta)(1 - q_0) + \theta(1 - q_1) - (\theta - p)^2}{m}.$$

Step 3: Unbiasedness of $\hat{\theta}_{\text{PPI}}$. By linearity of expectation,

$$\mathbb{E}[\hat{\theta}_{\text{PPI}}] = \mathbb{E}[\hat{\mu}] - \mathbb{E}[\hat{\Delta}] = p - (p - \theta) = \theta.$$

Step 4: Finite-sample variance of $\hat{\theta}_{\text{PPI}}$. Because the test and calibration samples are independent, $\hat{\mu}$ and $\hat{\Delta}$ are independent. Thus,

$$\text{Var}(\hat{\theta}_{\text{PPI}}) = \text{Var}(\hat{\mu} - \hat{\Delta}) = \text{Var}(\hat{\mu}) + \text{Var}(\hat{\Delta}) = \frac{p(1-p)}{n} + \frac{(1 - \theta)(1 - q_0) + \theta(1 - q_1) - (\theta - p)^2}{m}.$$

Step 5: Asymptotic normality. By the CLT,

$$\sqrt{n}(\hat{\mu} - p) \xrightarrow{d} \mathcal{N}(0, p(1-p)), \quad \sqrt{m}(\hat{\Delta} - (p - \theta)) \xrightarrow{d} \mathcal{N}(0, \text{Var}(\hat{Y} - Y)).$$

If $n, m \rightarrow \infty$ with $n/m \rightarrow \gamma_1 \in (0, \infty)$, then

$$\sqrt{n}(\hat{\Delta} - (p - \theta)) = \sqrt{\frac{n}{m}} \sqrt{m}(\hat{\Delta} - (p - \theta)) \xrightarrow{d} \mathcal{N}(0, \gamma_1 \text{Var}(\hat{Y} - Y)),$$

by Slutsky's theorem. Independence of the two samples implies the joint convergence with independent limits, and therefore

$$\sqrt{n}(\hat{\theta}_{\text{PPI}} - \theta) = \sqrt{n}(\hat{\mu} - p) - \sqrt{n}(\hat{\Delta} - (p - \theta)) \xrightarrow{d} \mathcal{N}(0, p(1-p) + \gamma_1 \text{Var}(\hat{Y} - Y)).$$

With algebra and the assumption that $\frac{N}{n} = \frac{m+n}{n} \rightarrow \frac{1+\gamma_1}{\gamma_1}$, we have that

$$V_{\text{PPI}} = \frac{1+\gamma_1}{\gamma_1} \left\{ p(1-p) + \gamma_1 \left[(1-\theta)(1-q_0) + \theta(1-q_1) - (\theta-p)^2 \right] \right\}.$$

□

B.3. Proof of Proposition 2.3

Proof. We can rewrite the PPI++ estimator with a fixed tuning parameter $\lambda \in \mathbb{R}$ as:

$$\hat{\theta}_{\text{PPI++}}(\lambda) = \frac{1}{m} \sum_{j=1}^m Y_j + \lambda \left(\frac{1}{n} \sum_{i=1}^n \hat{Y}_i - \frac{1}{m} \sum_{j=1}^m \hat{Y}_j \right) = \lambda \hat{p} + (\hat{\theta}_{\text{class}} - \lambda \bar{\hat{Y}}_{\text{cal}}),$$

where

$$\hat{\theta}_{\text{class}} = \frac{1}{m} \sum_{j=1}^m Y_j, \quad \bar{\hat{Y}}_{\text{cal}} = \frac{1}{m} \sum_{j=1}^m \hat{Y}_j, \quad \hat{p} = \frac{1}{n} \sum_{i=1}^n \hat{Y}_i.$$

Unbiasedness. Under the joint model (1), $\mathbb{E}[\hat{\theta}_{\text{class}}] = \mathbb{E}[Y] = \theta$ and $\mathbb{E}[\hat{p}] = \mathbb{E}[\bar{\hat{Y}}_{\text{cal}}] = \mathbb{E}[\hat{Y}] = p$. Hence,

$$\mathbb{E}[\hat{\theta}_{\text{PPI++}}(\lambda)] = \mathbb{E}[\hat{\theta}_{\text{class}}] + \lambda(\mathbb{E}[\hat{p}] - \mathbb{E}[\bar{\hat{Y}}_{\text{cal}}]) = \theta + \lambda(p - p) = \theta.$$

Finite-sample variance. By independence of the test and calibration samples, $\hat{p}$ is independent of $(\hat{\theta}_{\text{class}}, \bar{\hat{Y}}_{\text{cal}})$. Therefore, we have that

$$\text{Var}(\hat{\theta}_{\text{PPI++}}(\lambda)) = \text{Var}(\lambda \hat{p}) + \text{Var}(\hat{\theta}_{\text{class}} - \lambda \bar{\hat{Y}}_{\text{cal}}).$$

Moreover,

$$\text{Var}(\hat{p}) = \frac{1}{n} \text{Var}(\hat{Y}) = \frac{p(1-p)}{n}, \quad \text{Var}(\hat{\theta}_{\text{class}}) = \frac{1}{m} \text{Var}(Y) = \frac{\theta(1-\theta)}{m}, \quad \text{Var}(\bar{\hat{Y}}_{\text{cal}}) = \frac{1}{m} \text{Var}(\hat{Y}) = \frac{p(1-p)}{m}.$$

For the covariance term, we simplify it as

$$\text{Cov}(\hat{\theta}_{\text{class}}, \bar{\hat{Y}}_{\text{cal}}) = \text{Cov}\left(\frac{1}{m} \sum_{j=1}^m Y_j, \frac{1}{m} \sum_{j=1}^m \hat{Y}_j\right) = \frac{1}{m} \text{Cov}(Y, \hat{Y}),$$

since cross-sample terms vanish by independence across j . Also,

$$\text{Cov}(Y, \hat{Y}) = \mathbb{E}[Y\hat{Y}] - \mathbb{E}[Y]\mathbb{E}[\hat{Y}] = \Pr(Y=1, \hat{Y}=1) - \theta p = \theta q_1 - \theta p = \theta(q_1 - p).$$

Putting these together gives

$$\text{Var}(\hat{\theta}_{\text{PPI++}}(\lambda)) = \frac{\theta(1-\theta)}{m} + \lambda^2 \left(\frac{p(1-p)}{n} + \frac{p(1-p)}{m} \right) - \frac{2\lambda}{m} \theta(q_1 - p).$$

Asymptotic normality. By the CLT,

$$\sqrt{n}(\hat{p} - p) \xrightarrow{d} \mathcal{N}(0, p(1-p)).$$

For the calibration component, define i.i.d. variables

$$W_j = Y_j - \lambda \hat{Y}_j, \quad \mathbb{E}[W_j] = \theta - \lambda p, \quad \text{Var}(W_j) = \text{Var}(Y - \lambda \hat{Y}) = \theta(1-\theta) + \lambda^2 p(1-p) - 2\lambda \theta(q_1 - p).$$

Then

$$\hat{\theta}_{\text{class}} - \lambda \bar{Y}_{\text{cal}} = \frac{1}{m} \sum_{j=1}^m W_j,$$

and hence, again by the CLT,

$$\sqrt{m} \left\{ (\hat{\theta}_{\text{class}} - \lambda \bar{Y}_{\text{cal}}) - (\theta - \lambda p) \right\} \xrightarrow{d} \mathcal{N}(0, \text{Var}(Y - \lambda \hat{Y})).$$

If $n, m \rightarrow \infty$ with $n/m \rightarrow \gamma_1 \in (0, \infty)$, then $\sqrt{n} = \sqrt{n/m} \sqrt{m}$ and $\sqrt{n/m} \rightarrow \sqrt{\gamma_1}$, so by Slutsky's theorem,

$$\sqrt{n} \left\{ (\hat{\theta}_{\text{class}} - \lambda \bar{Y}_{\text{cal}}) - (\theta - \lambda p) \right\} \xrightarrow{d} \mathcal{N}(0, \gamma_1 \text{Var}(Y - \lambda \hat{Y})).$$

Finally, using independence between the test and calibration samples,

$$\begin{aligned} \sqrt{n}(\hat{\theta}_{\text{PPI++}}(\lambda) - \theta) &= \lambda \sqrt{n}(\hat{p} - p) + \sqrt{n} \left\{ (\hat{\theta}_{\text{class}} - \lambda \bar{Y}_{\text{cal}}) - (\theta - \lambda p) \right\} \\ &\xrightarrow{d} \mathcal{N}(0, \lambda^2 p(1-p) + \gamma_1 \text{Var}(Y - \lambda \hat{Y})), \end{aligned}$$

where

$$\begin{aligned} \text{Var}(Y - \lambda \hat{Y}) &= \text{Var}(Y) + \lambda^2 \text{Var}(\hat{Y}) - 2\lambda \text{Cov}(Y, \hat{Y}) \\ &= \theta(1-\theta) + \lambda^2 p(1-p) - 2\lambda \theta(q_1 - p). \end{aligned}$$

Plugging this into the asymptotic variance and the $\frac{N}{n}$ scaling gives us

$$\begin{aligned} V_{\text{PPI++}}(\lambda) &= \frac{1 + \gamma_1}{\gamma_1} \left\{ \lambda^2 p(1-p) + \gamma_1 \left[\theta(1-\theta) + \lambda^2 p(1-p) - 2\lambda \theta(q_1 - p) \right] \right\} \\ &= \frac{1 + \gamma_1}{\gamma_1} \left\{ \gamma_1 \theta(1-\theta) + \lambda^2 (1 + \gamma_1) p(1-p) - 2\lambda \gamma_1 \theta(q_1 - p) \right\}. \end{aligned}$$

□

B.4. Proof of Proposition 2.4

Proof. Write the combined log-likelihood as

$$\ell_{n,m}(\eta) = \sum_{i=1}^n \ell_{\text{test}}(\hat{Y}_i; \eta) + \sum_{j=1}^m \ell_{\text{cal}}(Y_j, \hat{Y}_j; \eta), \quad \eta = (\theta, q_0, q_1)^\top,$$

where the test sample contributes only $\hat{Y}$ and the calibration sample contributes $(Y, \hat{Y})$.

Step 1: Scores and Fisher information. For a test-sample observation $\hat{Y} \in \{0, 1\}$, the model implies

$$P(\hat{Y} = 1) = p(\eta) = (1 - \theta)(1 - q_0) + \theta q_1,$$

hence

$$\ell_{\text{test}}(\hat{Y}; \eta) = \hat{Y} \log p + (1 - \hat{Y}) \log(1 - p).$$

Let

$$g(\eta) = \nabla_{\eta} p(\eta) = (q_0 + q_1 - 1, -(1 - \theta), \theta)^\top.$$

Taking derivative with respect to η yields the score

$$\dot{\ell}_{\text{test}}(\hat{Y}; \eta) = \frac{\hat{Y} - p}{p(1-p)} g(\eta).$$

Since $\text{Var}(\hat{Y}) = p(1-p)$, the (per-observation) Fisher information from the test sample is

$$I_{\text{test}}(\eta) = \mathbb{E} \left[\dot{\ell}_{\text{test}}(\hat{Y}; \eta) \dot{\ell}_{\text{test}}(\hat{Y}; \eta)^\top \right] = \frac{1}{p(1-p)} g(\eta) g(\eta)^\top,$$

which matches the stated matrix.

For a calibration-sample observation $(Y, \hat{Y})$, the complete-data likelihood factorizes as

$$P(Y = y, \hat{Y} = \hat{y}) = P(Y = y) P(\hat{Y} = \hat{y} \mid Y = y),$$

where $P(Y = 1) = \theta$, $P(\hat{Y} = 1 \mid Y = 0) = 1 - q_0$, and $P(\hat{Y} = 1 \mid Y = 1) = q_1$. Thus

$$\begin{aligned} \ell_{\text{cal}}(Y, \hat{Y}; \eta) &= Y \log \theta + (1 - Y) \log(1 - \theta) \\ &\quad + (1 - Y) \left[\hat{Y} \log(1 - q_0) + (1 - \hat{Y}) \log q_0 \right] + Y \left[\hat{Y} \log q_1 + (1 - \hat{Y}) \log(1 - q_1) \right]. \end{aligned}$$

The score components are

$$\dot{\ell}_\theta = \frac{Y - \theta}{\theta(1-\theta)}, \quad \dot{\ell}_{q_0} = (1 - Y) \left(\frac{1 - \hat{Y}}{q_0} - \frac{\hat{Y}}{1 - q_0} \right), \quad \dot{\ell}_{q_1} = Y \left(\frac{\hat{Y}}{q_1} - \frac{1 - \hat{Y}}{1 - q_1} \right).$$

We first compute the off-diagonal entries in the information matrix.

Cross-moment $\mathbb{E}[\dot{\ell}_\theta \dot{\ell}_{q_0}]$: $\dot{\ell}_{q_0}$ equals 0 when $Y = 1$. Using iterated expectations, we have that

$$\mathbb{E}[\dot{\ell}_\theta \dot{\ell}_{q_0}] = \mathbb{E}[\mathbb{E}[\dot{\ell}_\theta \dot{\ell}_{q_0} \mid Y]] = (1 - \theta) \mathbb{E}[\dot{\ell}_\theta \dot{\ell}_{q_0} \mid Y = 0].$$

Given $Y = 0$, $\dot{\ell}_\theta = \frac{-\theta}{\theta(1-\theta)} = \frac{-1}{1-\theta}$, so:

$$\mathbb{E}[\dot{\ell}_\theta \dot{\ell}_{q_0} \mid Y = 0] = \frac{-1}{1-\theta} \mathbb{E}[\dot{\ell}_{q_0} \mid Y = 0] = \frac{-1}{1-\theta} \times 0 = 0.$$

Cross-moment $\mathbb{E}[\dot{\ell}_\theta \dot{\ell}_{q_1}]$: First note that $\dot{\ell}_{q_1} = 0$ when $Y = 0$, therefore we have that

$$\mathbb{E}[\dot{\ell}_\theta \dot{\ell}_{q_1}] = \theta \mathbb{E}[\dot{\ell}_\theta \dot{\ell}_{q_1} \mid Y = 1].$$

Given $Y = 1$: $\dot{\ell}_\theta = \frac{1}{\theta}$, we conclude that

$$\mathbb{E}[\dot{\ell}_\theta \dot{\ell}_{q_1} \mid Y = 1] = \frac{1}{\theta} \mathbb{E}[\dot{\ell}_{q_1} \mid Y = 1] = \frac{1}{\theta} \times 0 = 0.$$

Cross-moment $\mathbb{E}[\dot{\ell}_{q_0} \dot{\ell}_{q_1}]$: $\dot{\ell}_{q_0} \dot{\ell}_{q_1} = (1 - Y) \cdot Y \cdot (\dots)$; since $Y \in \{0, 1\}$ implies $(1 - Y) \cdot Y = 0$ always.

Therefore, the calibration Fisher information is diagonal with entries

$$I_{\text{cal}}(\eta) = \text{diag} \left(\mathbb{E}[\dot{\ell}_\theta^2], \mathbb{E}[\dot{\ell}_{q_0}^2], \mathbb{E}[\dot{\ell}_{q_1}^2] \right) = \text{diag} \left(\frac{1}{\theta(1-\theta)}, \frac{1-\theta}{q_0(1-q_0)}, \frac{\theta}{q_1(1-q_1)} \right).$$

Step 2: Asymptotic normality of the joint MLE. Let $N = n + m$ and write the full score as

$$\dot{\ell}_{n,m}(\eta) = \sum_{i=1}^n \dot{\ell}_{\text{test}}(\hat{Y}_i; \eta) + \sum_{j=1}^m \dot{\ell}_{\text{cal}}(Y_j, \hat{Y}_j; \eta).$$

By construction, $\mathbb{E}[\dot{\ell}_{n,m}(\eta)] = 0$, and by independence,

$$\text{Var}\left(\frac{1}{\sqrt{N}}\dot{\ell}_{n,m}(\eta)\right) = \frac{n}{N} I_{\text{test}}(\eta) + \frac{m}{N} I_{\text{cal}}(\eta) \rightarrow \frac{\gamma_1}{1 + \gamma_1} I_{\text{test}}(\eta) + \frac{1}{1 + \gamma_1} I_{\text{cal}}(\eta) =: I_{\gamma_1}(\eta),$$

since $n/m \rightarrow \gamma_1 \in (0, \infty)$ implies $n/N \rightarrow \gamma_1/(1 + \gamma_1)$ and $m/N \rightarrow 1/(1 + \gamma_1)$. A multivariate CLT therefore gives

$$\frac{1}{\sqrt{N}}\dot{\ell}_{n,m}(\eta) \xrightarrow{d} \mathcal{N}(0, I_{\gamma_1}(\eta)).$$

Under standard regularity conditions, we have that

$$\sqrt{N}(\hat{\eta}_{\text{MLE}} - \eta) = \left(-\frac{1}{N}\ddot{\ell}_{n,m}(\tilde{\eta})\right)^{-1} \left(\frac{1}{\sqrt{N}}\dot{\ell}_{n,m}(\eta)\right) \xrightarrow{d} \mathcal{N}(0, I_{\gamma_1}(\eta)^{-1}).$$

Step 3: Closed form for the $(1, 1)$ entry. Write $w = \gamma_1/(1 + \gamma_1)$ and note that

$$I_{\gamma_1}(\eta) = w I_{\text{test}}(\eta) + \frac{1}{1 + \gamma_1} I_{\text{cal}}(\eta) = D + uu^\top,$$

where

$$D = \frac{1}{1 + \gamma_1} I_{\text{cal}}(\eta) \quad (\text{diagonal}), \quad u = \sqrt{\frac{w}{p(1-p)}} g(\eta).$$

Since D is invertible (parameters in the interior), the Sherman–Morrison formula gives

$$(D + uu^\top)^{-1} = D^{-1} - \frac{D^{-1}uu^\top D^{-1}}{1 + u^\top D^{-1}u}.$$

Here,

$$D^{-1} = (1 + \gamma_1) \text{diag}\left(\theta(1 - \theta), \frac{q_0(1 - q_0)}{1 - \theta}, \frac{q_1(1 - q_1)}{\theta}\right),$$

and a direct computation shows

$$u^\top D^{-1}u = \frac{\gamma_1}{p(1-p)} \left[(q_0 + q_1 - 1)^2 \theta(1 - \theta) + (1 - \theta)q_0(1 - q_0) + \theta q_1(1 - q_1) \right].$$

Moreover, the first component satisfies

$$(D^{-1}u)_1^2 = \frac{\gamma_1(1 + \gamma_1)\theta^2(1 - \theta)^2(q_0 + q_1 - 1)^2}{p(1-p)}.$$

Substituting into the Sherman–Morrison expression and simplifying yields

$$[I_{\gamma_1}(\eta)^{-1}]_{11} = (1 + \gamma_1)\theta(1 - \theta) \frac{p(1-p) + \gamma_1 \left[(1 - \theta)q_0(1 - q_0) + \theta q_1(1 - q_1) \right]}{p(1-p) + \gamma_1 \left[(q_0 + q_1 - 1)^2 \theta(1 - \theta) + (1 - \theta)q_0(1 - q_0) + \theta q_1(1 - q_1) \right]},$$

which is the stated closed form. The marginal asymptotic normality of $\hat{\theta}_{\text{MLE}}$ follows by taking the first coordinate of the multivariate limit. $\square$

B.5. Proof of Proposition 3.1

Proof. Recall the observed data are $O = (R, RY, \hat{Y})$, where $R \in \{0, 1\}$ indicates whether Y is observed and the labeling mechanism satisfies $P(R = 1 \mid Y, \hat{Y}) = P(R = 1) \in (0, 1)$.

Define

$$\mu(\hat{Y}) = \mathbb{E}[Y \mid \hat{Y}], \quad \theta = \mathbb{E}[Y] = \mathbb{E}[\mu(\hat{Y})].$$

We derive the efficient influence function by identifying the *canonical gradient* of θ in the observed-data model.

Step 1: Tangent space representation. Let $\{P_\varepsilon : \varepsilon \in (-\delta, \delta)\}$ be any regular parametric submodel through the true distribution P_0 (with $P_{\varepsilon=0} = P_0$). Under $P(R = 1 \mid Y, \hat{Y}) = \pi$, the observed-data density factorizes as

$$p_\varepsilon(o) = p_{\varepsilon,R}(r) p_{\varepsilon,\hat{Y}}(\hat{y}) \{p_{\varepsilon,Y|\hat{Y}}(y \mid \hat{y})\}^r,$$

so the observed-data score admits the decomposition

$$s_\varepsilon(O) = s_R(R) + s_{\hat{Y}}(\hat{Y}) + R s_{Y|\hat{Y}}(Y, \hat{Y}), \quad (19)$$

where the components satisfy the usual mean-zero constraints

$$\mathbb{E}[s_R(R)] = 0, \quad \mathbb{E}[s_{\hat{Y}}(\hat{Y})] = 0, \quad \mathbb{E}[s_{Y|\hat{Y}}(Y, \hat{Y}) \mid \hat{Y}] = 0.$$

Thus the (closure of the) observed-data tangent space consists of all functions of the form (19).

Step 2: Pathwise derivative of θ . Along the submodel, $\theta(\varepsilon) = \mathbb{E}_\varepsilon[Y] = \mathbb{E}_\varepsilon[\mu_\varepsilon(\hat{Y})]$, where $\mu_\varepsilon(\hat{y}) = \mathbb{E}_\varepsilon[Y \mid \hat{Y} = \hat{y}]$. Differentiating at $\varepsilon = 0$ and using standard score calculus gives

$$\left. \frac{d}{d\varepsilon} \theta(\varepsilon) \right|_{\varepsilon=0} = \mathbb{E}[\{\mu(\hat{Y}) - \theta\} s_{\hat{Y}}(\hat{Y})] + \mathbb{E}[(Y - \mu(\hat{Y})) s_{Y|\hat{Y}}(Y, \hat{Y})]. \quad (20)$$

There is no contribution from $s_R(R)$ because θ depends only on the marginal law of $(Y, \hat{Y})$.

Step 3: Candidate influence function and verification. Consider

$$\phi^*(O) = \mu(\hat{Y}) - \theta + \frac{R}{\pi} \{Y - \mu(\hat{Y})\}. \quad (21)$$

First, $\mathbb{E}[\phi^*(O)] = 0$ since $\mathbb{E}[\mu(\hat{Y})] = \theta$ and

$$\mathbb{E}\left[\frac{R}{\pi} \{Y - \mu(\hat{Y})\} \mid \hat{Y}\right] = \frac{\mathbb{E}[R \mid \hat{Y}]}{\pi} \mathbb{E}[Y - \mu(\hat{Y}) \mid \hat{Y}] = 0.$$

Next, we show that ϕ^* represents the pathwise derivative: for any score $s(O)$ of the form (19),

$$\mathbb{E}[\phi^*(O) s(O)] = \mathbb{E}[\phi^*(O) s_R(R)] + \mathbb{E}[\phi^*(O) s_{\hat{Y}}(\hat{Y})] + \mathbb{E}[\phi^*(O) R s_{Y|\hat{Y}}(Y, \hat{Y})].$$

The first term is zero because $\mathbb{E}[\phi^*(O) \mid R] = 0$ (by the same calculation as above) and $\mathbb{E}[s_R(R)] = 0$. For the second term, the residual part drops out:

$$\mathbb{E}\left[\frac{R}{\pi} \{Y - \mu(\hat{Y})\} s_{\hat{Y}}(\hat{Y})\right] = \mathbb{E}\left[s_{\hat{Y}}(\hat{Y}) \mathbb{E}\left[\frac{R}{\pi} \{Y - \mu(\hat{Y})\} \mid \hat{Y}\right]\right] = 0,$$

so

$$\mathbb{E}[\phi^*(O) s_{\hat{Y}}(\hat{Y})] = \mathbb{E}[\{\mu(\hat{Y}) - \theta\} s_{\hat{Y}}(\hat{Y})].$$

For the third term, note that $\mathbb{E}[s_{Y|\hat{Y}}(Y, \hat{Y}) \mid \hat{Y}] = 0$ implies $\mathbb{E}[R s_{Y|\hat{Y}}(Y, \hat{Y}) \mid \hat{Y}] = \pi \cdot 0 = 0$, hence

$$\mathbb{E}[\{\mu(\hat{Y}) - \theta\} R s_{Y|\hat{Y}}(Y, \hat{Y})] = 0,$$

and therefore

$$\mathbb{E}[\phi^*(O) R s_{Y|\hat{Y}}(Y, \hat{Y})] = \mathbb{E}\left[\frac{R}{\pi}\{Y - \mu(\hat{Y})\} R s_{Y|\hat{Y}}(Y, \hat{Y})\right] = \mathbb{E}[(Y - \mu(\hat{Y})) s_{Y|\hat{Y}}(Y, \hat{Y})],$$

using $R^2 = R$ and $\mathbb{E}[Rf(Y, \hat{Y})] = \pi \mathbb{E}[f(Y, \hat{Y})]$ under $P(R = 1 \mid Y, \hat{Y}) = \pi$.

Combining the pieces yields

$$\mathbb{E}[\phi^*(O) s(O)] = \mathbb{E}[\{\mu(\hat{Y}) - \theta\} s_{\hat{Y}}(\hat{Y})] + \mathbb{E}[(Y - \mu(\hat{Y})) s_{Y|\hat{Y}}(Y, \hat{Y})],$$

which matches the pathwise derivative in (20). Hence ϕ^* is the canonical gradient of θ in the observed-data model, and therefore the efficient influence function. This is exactly (12). $\square$

B.6. Proof of Proposition 3.2

Proof. Step 1: Derivation of the estimator. By Proposition 3.1, an efficient influence function for θ is

$$\phi^*(R, Y, \hat{Y}) = \mu(\hat{Y}) - \theta + \frac{R}{P(R = 1)}\{Y - \mu(\hat{Y})\}.$$

The standard EIF (one-step) estimator is obtained by solving the empirical estimating equation with μ replaced by $\hat{\mu}$ and $P(R = 1)$ replaced by $\hat{P}(R = 1)$:

$$0 = \frac{1}{N} \sum_{i=1}^N \phi^*(R_i, Y_i, \hat{Y}_i; \theta, \hat{\mu}, \hat{P}(R = 1)).$$

Plugging in ϕ^* and expanding gives

$$0 = \frac{1}{n+m} \sum_{i=1}^{n+m} \left[\hat{\mu}(\hat{Y}_i) - \theta + \frac{R_i}{\hat{P}(R = 1)}\{Y_i - \hat{\mu}(\hat{Y}_i)\} \right].$$

Rearranging for θ yields:

$$\theta = \frac{1}{N} \sum_{i=1}^N \hat{\mu}(\hat{Y}_i) + \frac{1}{N} \sum_{i=1}^N \frac{R_i}{\hat{P}(R = 1)}\{Y_i - \hat{\mu}(\hat{Y}_i)\}.$$

Finally, under our setup $P(R = 1) = m/(n+m)$, so plugging that as our estimate gives us (13).

Step 2: Asymptotic normality. By standard M-estimation theory for one-step estimators (Bickel et al., 1993),

$$\sqrt{N}(\hat{\theta}_{\text{EIF}} - \theta) = \frac{1}{\sqrt{N}} \sum_{i=1}^N \phi^*(R_i, Y_i, \hat{Y}_i) + o_p(1).$$

The central limit theorem then gives $\sqrt{N}(\hat{\theta}_{\text{EIF}} - \theta) \xrightarrow{d} \mathcal{N}(0, V_{\text{EIF}})$ where $V_{\text{EIF}} = \text{Var}(\phi^*)$.

Step 3: Variance calculation. Write $\phi^* = A + B$ where $A = \mu(\hat{Y}) - \theta$ and $B = (R/P(R = 1))(Y - \mu(\hat{Y}))$. Since $\mathbb{E}[Y - \mu(\hat{Y}) \mid \hat{Y}] = 0$, we have $\text{Cov}(A, B) = 0$, so

$$V_{\text{EIF}} = \text{Var}(A) + \text{Var}(B).$$

For the first term:

$$\text{Var}(A) = \text{Var}(\mu(\hat{Y})) = p(1-p)(\mu(1) - \mu(0))^2.$$

For the second term, using $R^2 = R$ and independence of R from $(Y, \hat{Y})$:

$$\text{Var}(B) = \frac{1}{P(R=1)^2} \mathbb{E}[R(Y - \mu(\hat{Y}))^2] = \frac{1}{P(R=1)} \mathbb{E}[\text{Var}(Y | \hat{Y})].$$

Since $P(R=1) = m/N = 1/(1 + \gamma_1)$ where $\gamma_1 = n/m$, we have

$$\text{Var}(B) = (1 + \gamma_1)[(1-p)\mu(0)(1-\mu(0)) + p\mu(1)(1-\mu(1))].$$

Step 4: Simplification for binary case. Substituting $\mu(1) = q_1\theta/p$, $\mu(0) = (1-q_1)\theta/(1-p)$, and $p = q_1\theta + (1-q_0)(1-\theta)$, we obtain after algebra:

$$\begin{aligned} \mu(1) - \mu(0) &= \frac{\theta(1-\theta)(q_0 + q_1 - 1)}{p(1-p)}, \\ (1-p)\mu(0)(1-\mu(0)) + p\mu(1)(1-\mu(1)) &= \frac{\theta(1-\theta)[q_0(1-q_0)(1-\theta) + q_1(1-q_1)\theta]}{p(1-p)}. \end{aligned}$$

Combining the two pieces completes the proof. $\square$

B.7. Asymptotic equivalence of V_{MLE} and V_{EIF}

Proposition B.1. *Under the model (1) with $\gamma_1 = n/m$, the asymptotic variance of the MLE equals the asymptotic variance of the EIF estimator:*

$$V_{\text{MLE}} := [I_{\gamma_1}(\eta)^{-1}]_{11} = V_{\text{EIF}}.$$

Proof. Define

$$\begin{aligned} C &:= (q_0 + q_1 - 1)^2\theta(1-\theta), \\ B &:= (1-\theta)q_0(1-q_0) + \theta q_1(1-q_1). \end{aligned}$$

Key identity. By the law of total variance applied to $\hat{Y}$:

$$p(1-p) = \text{Var}(\hat{Y}) = \underbrace{\text{Var}(\mathbb{E}[\hat{Y} | Y])}_{=C} + \underbrace{\mathbb{E}[\text{Var}(\hat{Y} | Y)]}_{=B} = C + B.$$

Simplifying V_{MLE} . From Proposition 2.4, the $(1, 1)$ entry of the inverse Fisher information is

$$V_{\text{MLE}} = (1 + \gamma_1)\theta(1-\theta) \frac{p(1-p) + \gamma_1 B}{p(1-p) + \gamma_1(C+B)}.$$

Substituting $p(1-p) = C + B$ in the denominator:

$$\begin{aligned} V_{\text{MLE}} &= (1 + \gamma_1)\theta(1-\theta) \frac{(C+B) + \gamma_1 B}{(C+B) + \gamma_1(C+B)} \\ &= (1 + \gamma_1)\theta(1-\theta) \frac{C + (1 + \gamma_1)B}{(1 + \gamma_1)(C+B)} \\ &= \frac{\theta(1-\theta)}{p(1-p)} [C + (1 + \gamma_1)B]. \end{aligned}$$

Comparing with V_{EIF} . From Proposition 3.2:

$$V_{\text{EIF}} = \frac{\theta(1-\theta)}{p(1-p)} [\theta(1-\theta)(q_0 + q_1 - 1)^2 + (1 + \gamma_1)(q_0(1-q_0)(1-\theta) + q_1(1-q_1)\theta)].$$

This is precisely $\frac{\theta(1-\theta)}{p(1-p)} [C + (1 + \gamma_1)B]$, which concludes the proof. $\square$

B.8. Proof of Proposition 3.3

Proof. Let $\kappa = q_0 + q_1 - 1 \in (0, 1]$, with $\kappa = 1$ if and only if $q_0 = q_1 = 1$ (perfect classification).

We introduce the following notation for key quantities:

$$\begin{aligned} V_1 &= p(1-p) = \text{Var}(\hat{Y}), \\ V_2 &= (1-\theta)(1-q_0) + \theta(1-q_1) - (\theta-p)^2 = \text{Var}(Y - \hat{Y}), \\ V_3 &= (1-\theta)q_0(1-q_0) + \theta q_1(1-q_1) = \mathbb{E}[\text{Var}(\hat{Y}|Y)]. \end{aligned}$$

Here V_2 is the variance of $\delta = Y - \hat{Y}$, noting that $\mathbb{E}[\delta] = \theta - p$ and $\mathbb{E}[\delta^2] = \mathbb{P}(Y \neq \hat{Y}) = (1-\theta)(1-q_0) + \theta(1-q_1)$.

We further divide both by a positive factor of $\frac{1+\gamma_1}{\gamma_1}$, which gives us

$$\frac{\gamma_1}{1+\gamma_1} V_{\text{PPI}} = V_1 + \gamma_1 V_2, \quad \frac{\gamma_1}{1+\gamma_1} V_{\text{RG}} = \frac{1}{\kappa^2} (V_1 + \gamma_1 V_3).$$

Step 1: Decompose the difference. We have that

$$\begin{aligned} \frac{\gamma_1}{1+\gamma_1} (V_{\text{RG}} - V_{\text{PPI}}) &= \frac{V_1 + \gamma_1 V_3}{\kappa^2} - V_1 - \gamma_1 V_2 \\ &= V_1 \left(\frac{1}{\kappa^2} - 1 \right) + \gamma_1 \left(\frac{V_3}{\kappa^2} - V_2 \right) \\ &= \frac{(1-\kappa^2)V_1}{\kappa^2} + \frac{\gamma_1(V_3 - \kappa^2 V_2)}{\kappa^2} \\ &= \frac{(1-\kappa)(1+\kappa)V_1}{\kappa^2} + \frac{\gamma_1(V_3 - \kappa^2 V_2)}{\kappa^2}. \end{aligned}$$

Step 2: First term is non-negative. Since $0 < \kappa \leq 1$, we have $(1-\kappa) \geq 0$, $(1+\kappa) > 0$, and $V_1 = p(1-p) \geq 0$. Thus

$$\frac{(1-\kappa)(1+\kappa)V_1}{\kappa^2} \geq 0,$$

with equality if and only if $\kappa = 1$ or $p \in \{0, 1\}$.

Step 3: Second term is non-negative. It remains to show that $V_3 - \kappa^2 V_2 \geq 0$. By the key identity in Section B.7, we have

$$V_3 = V_1 - \theta(1-\theta)\kappa^2.$$

Thus:

$$V_3 - \kappa^2 V_2 = V_1 - \theta(1-\theta)\kappa^2 - \kappa^2 V_2 = V_1 - \kappa^2 (V_2 + \theta(1-\theta)).$$

We claim this expression equals $(1-\kappa)Q$ for some $Q \geq 0$. Algebraic manipulation (verified symbolically) shows:

$$V_3 - \kappa^2 V_2 = (1-\kappa) \cdot Q(\theta, q_0, q_1),$$

where

$$\begin{aligned} Q(\theta, q_0, q_1) &= (1+\kappa)V_3 - \kappa^2(1-\kappa)\theta(1-\theta) \\ &= (q_0 + q_1) \left((1-\theta)q_0(1-q_0) + \theta q_1(1-q_1) \right) - (q_0 + q_1 - 1)^2 (2 - q_0 - q_1) \theta (1-\theta). \end{aligned}$$

Moreover, $Q(\theta, q_0, q_1) \geq 0$ for all $\theta \in (0, 1)$ and $q_0, q_1 \in (0, 1)$ with $q_0 + q_1 > 1$.

Since $(1-\kappa) = 2 - q_0 - q_1 \geq 0$ (as $q_0, q_1 < 1$) and $Q \geq 0$, we have:

$$V_3 - \kappa^2 V_2 \geq 0,$$

with equality if and only if $\kappa = 1$ (i.e., $q_0 = q_1 = 1$).

Step 4: Conclusion. Combining Steps 2 and 3:

$$\frac{\gamma_1}{1 + \gamma_1} (V_{\text{RG}} - V_{\text{PPI}}) = \frac{(1 - \kappa)(1 + \kappa)V_1}{\kappa^2} + \frac{\gamma_1(V_3 - \kappa^2 V_2)}{\kappa^2} \geq 0.$$

Therefore $V_{\text{PPI}} \leq V_{\text{RG}}$.

Equality condition. Equality holds if and only if both terms vanish. The first term vanishes when $\kappa = 1$ or $p \in \{0, 1\}$. The second term vanishes when $\kappa = 1$. Since $p = (1 - \theta)(1 - q_0) + \theta q_1$ and $\theta, q_0, q_1 \in (0, 1)$, we have $p \in (0, 1)$. Thus equality requires $\kappa = 1$, which means $q_0 + q_1 = 2$, i.e., $q_0 = q_1 = 1$ (the classifier is perfect). $\square$

B.9. Calibration fits for results in Section 5.5

We visualize the calibration functions $\mu(\hat{y}) = \mathbb{E}[Y \mid \hat{Y} = \hat{y}]$ estimated by each method in Figure 8. The true conditional expectation (dashed black line) is determined by the simulation DGP: since $Y \mid Z \sim \mathcal{N}(\mu_Z, \sigma^2)$ and $\hat{Y} = Z$, the calibration function is piecewise constant with jumps at $\hat{Y} \in \{1, 2, 3\}$. Spline/GAM methods closely approximate this step function, while linear calibration imposes a restrictive functional form that introduces bias when the true relationship is non-linear. This explains the wider confidence intervals observed for the linear EIF estimator.

Y vs $\hat{Y}$ with Calibration Fits (Discrete Predictions)

DGP: $Z \sim \text{Unif}\{1, 2, 3\}$, $Y|Z \sim \mathcal{N}(\mu_Z, 1)$, $\hat{Y} = Z$. Dashed line = perfect calibration.

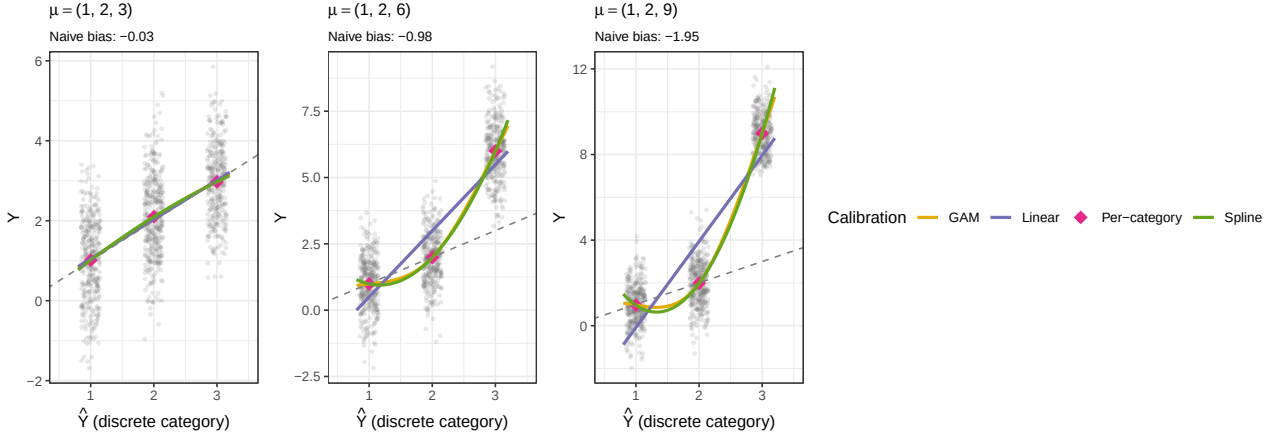


Figure 8. Calibration fits for the continuous Y simulation.